

IEMA: Exploring Complex Data Spaces with Body-Motion Interaction

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Spaces with Body-Motion
Interaction

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Abstract

Interactive search in complex data spaces is a crucial and challenging task in audio data retrieval field, where efficient methods are needed to recommend users relevant content from vast audio databases. This task can be addressed through various approaches, often combining multiple techniques together. This thesis proposes the Immersive Evolutionary Motion-controlled Algorithm (IEMA), an interactive system that combines Interactive Evolutionary Algorithms and body motion interaction to guide the exploration of discrete audio data. To aid in the design of such systems, it also presents an evaluation framework that informs how to develop and assess similar interactive evolutionary systems. Through stage-wise prototype design and experimentation, the thesis demonstrates the applicability of IEMA for high-dimensional discrete audio search. The thesis also shares the software implementation of IEMA as open-source. Furthermore, it highlights potential areas for future research in the field of interactive evolutionary music systems.

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Chapter 1

Introduction

Since the advent of computers, interactive systems have been a central focus of research, driven by the desire for more engaging and expressive ways to interact with technology. This has led to the development of sophisticated methods such as gesture-based control, spatial tracking, and adaptive generative systems. These advancements have opened new possibilities for artistic exploration, particularly in interactive music, where users can engage with sound in more immersive and dynamic ways.

The initial motivation for this thesis emerged from an earlier project focused on building an interactive, motion-controlled music system. This early prototype was a body-driven voice processor, where vocal input was processed using various audio effects. The parameters controlling these effects were controlled by the body, using Motion Capture (mocap) technology. This allowed users to shape the vocal input dynamically through physical movement in real-time. While this prototype demonstrated the potential of motion-controlled audio processing, the direct motion-to-sound mappings often felt unintuitive and repetitive over time. It raised further questions about adaptability: could the system evolve based on the user's preferences rather than relying solely on predefined mappings? This question led to the next stage of development, which introduced and assessed adaptive mechanisms to refine the interaction.

Designing systems that can adapt to individual preferences is one of the powerful approaches to creating interactions that feel intuitive, engaging, and responsive (Perry Cook 2017). This perspective formed a turning point in the direction of the research by highlighting adaptability not just as a system feature, but as a necessary quality for navigating complex and dynamic media environments. But how can a system be made adaptive, and how can users meaningfully interact with large and evolving data? There are many possible approaches, depending on how the system is structured, what data it handles, and how the interaction is designed (Turchet 2019).

Some of the most effective strategies for designing interactive and adaptive systems are those inspired by nature. From neural networks modeled after the human brain to swarm intelligence based on collective animal behavior, biological processes have served as powerful blueprints for problem-solving in artificial systems (Cui et al. 2024; SS and HS 2022).

In interactive systems, where outcomes are shaped by user preferences, evolution alone may be insufficient. Richard Dawkins' work on biomorphs introduced

the concept of Interactive Evolutionary Computation (IEC), emphasizing nature as an ever-evolving, interactive system of information and proposing the involvement of the user in the evolutionary process, known as Interactive Evolutionary Algorithms (IEA) (Dawkins 1986). IEAs build upon traditional evolutionary algorithms but incorporate human subjective feedback into the process. This allows users to directly influence evolution. This makes IEAs particularly effective for creative applications, where outcomes are driven by user interaction and continuous adaptation to individual preferences (Bäck 2018).

In most evolutionary algorithms, the focus is often on continuous data. This approach works well for optimization tasks where the solution space can be refined gradually. However, in many real-world applications, especially in creative fields like music, sound design, and other artistic domains, the data is inherently discrete. Evolving in such discrete spaces presents unique challenges, as the evolutionary process must consider the boundaries between distinct entities rather than gradient transitions. These challenges make it an interesting avenue for exploration, as they require novel adaptations of traditional evolutionary algorithms and body-motion interaction to effectively navigate and evolve discrete solution spaces.

In light of these considerations, the IEMI system, which serves as the preliminary work for this thesis, explored the use of Interactive Evolutionary Algorithms (IEA) to adaptively evolve dynamic soundscapes (Memis and Kelkar 2024). It compared both continuous and discrete data evolution, examining how real-time audio effects could be adapted based on user input, as well as how audio samples from a large dataset could be evolved and selected to fit user preferences. In this system, users interacted with the soundscape through a motion-capture setup and provided real-time feedback. The results of this study demonstrated the potential of IEAs for user-driven exploration of discrete data spaces, and highlighted the challenges in evaluating such systems. However, it lacked a systematic algorithmic implementation and did not provide a clear design framework or assessment metrics for guiding the development of such systems. Additionally, the study primarily focused on quantitative user assessments.

With all these considerations in mind, this thesis introduces *IEMA*, an interactive evolutionary system designed to explore and evolve dynamic soundscapes based on user preferences. *IEMA* combines an IEA with discrete data space and user interaction to continuously adapt a population of audio samples towards a desired artificial sound scape.

The following three components form the backbone of *IEMA*:

- **Interactive Evolutionary Algorithm:** is responsible for the evolutionary operations to evolve the point cloud in the room towards user-preferred audio characteristics.
- **Dataset (Corpus):** serves as the search space for the algorithm, containing all available raw audio samples along with their extracted features.
- **Spatial Interaction Framework:** enables real-time interaction with the population by projecting high-dimensional individuals into a three-dimensional space, utilizing motion capture system, and acquiring feedback from the user.

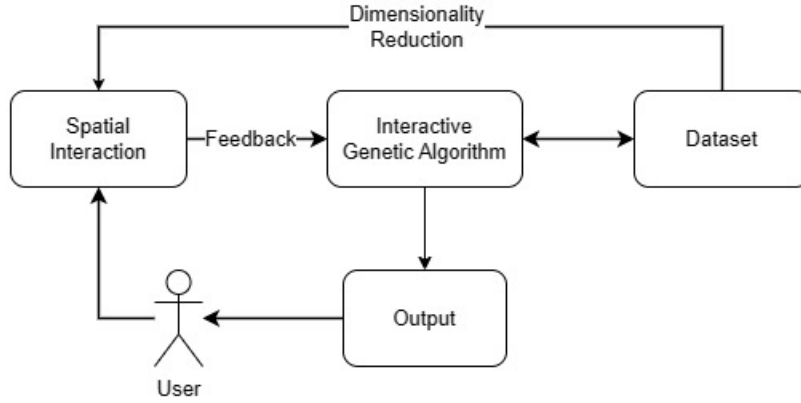


Figure 1.1: Main components of the proposed system.

Figure 1.1 illustrates these key components and their interactions within the system.

Moreover, the evaluation of systems like IEMA presents significant challenges. Therefore, it is difficult to know how to design such systems. In IEMA, the optimization process is guided by subjective human feedback rather than predefined fitness functions, and this makes conventional EA evaluation methods insufficient. Additionally, when working with discrete data instead of continuous, the lack of smooth gradients further complicates the analysis. These highlight the need for specialized evaluation techniques to better understand how such a system operates and evolves over time, therefore inform the design and implementation of IEMA.

To address this, this thesis follows an iterative process of implementation and evaluation, refining both the system design and the computational metrics used to assess it. By integrating evaluation throughout the development cycle, the research aims to establish a clearer understanding of how such systems function and how they can be improved. The work is built on the earlier explorations in the IEMI project, which compared interaction with continuous and discrete data. From there, an initial prototype (P1) was developed and evaluated, which explored and also highlighted the limitations of both the algorithm and evaluation metrics. This led to the final prototype (P2), a larger-scale implementation that focused on discrete data and introduced more comprehensive evaluation tools. Figure 1.2 illustrates this stage-wise development flow.

Ultimately, this research makes novel contributions by applying IEAs to explore discrete data, integrating motion capture for immersive, body-based interaction, utilizing a unary feedback method for user input, and implementing an open-source software system that runs all these components in parallel. Figure 1.3 illustrates an example of the IEMA interaction setup within the interactive room, showcasing how the user’s movements, captured by the motion capture system, contribute to the evolutionary process.

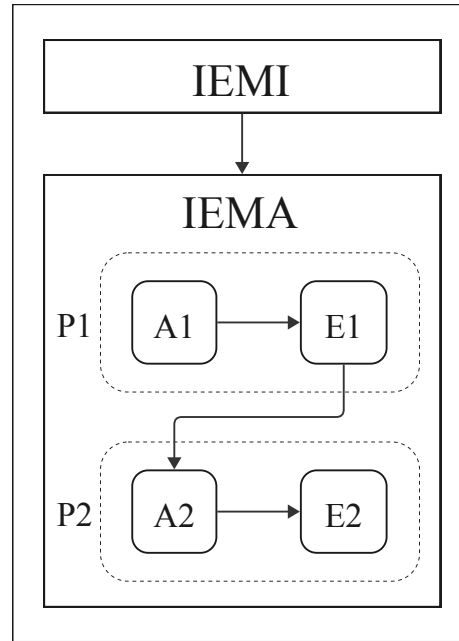


Figure 1.2: Structure of the prototype-driven methodology. The work builds on the IEMI project and proceeds through two development stages within IEMA. In Prototype 1 (P1), the initial implementation (A1) is evaluated by Initial Evaluation (E1). The results of E1 then inform the development of the refined implementation (A2) in Prototype 2 (P2), which is again evaluated by the Final Evaluation (E2). Each stage informs the next through iterative feedback.

1.1 Research Questions & Outcomes

This thesis aims to explore interactive music systems that utilize evolutionary algorithms, while also investigating methods for evaluating these algorithms to inform its design and implementation.

To achieve this, the following research questions are asked:

1. How can an interactive music system be designed to facilitate three-dimensional navigation and feedback-driven exploration of a high-dimensional discrete space, where sonic material evolves?
2. How can an algorithm be designed and implemented to facilitate the interactive evolution of discrete audio data using subjective user feedback?
3. What evaluation metrics can be developed and applied to analyze the behavior of the aforementioned system, and how can they provide insight into its capabilities?

Given the research questions, the main outcomes of this work are:

1. Implementation of interaction and feedback mechanisms for engaging with an evolutionary algorithm, enabling discrete data search and real-time playback.
2. Algorithmic implementation of evolutionary algorithm for high-dimensional discrete data exploration with user's unary feedback.

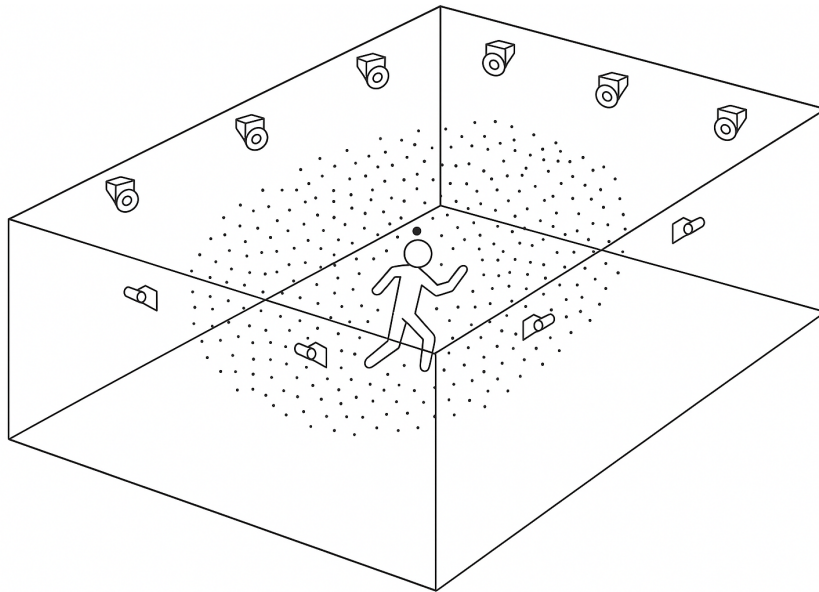


Figure 1.3: An example illustration of a user interacting with the IEMA. (The sizes and number of speakers and cameras are symbolic and do not represent the actual setup.)

3. Development and application of evaluation metrics to analyze and assess the system's behavior and outcomes.
4. Open-source code for further research and development.¹

This thesis is organized into 6 chapters. The Introduction outlines the topic, motivation, scope, and research questions. The Background reviews relevant literature as well as related and preliminary work. The Methodology explains the implementation of the system and the evaluation metrics, and the Final System Description follows with the final configuration of the IEMA. The Results & Discussion presents results and findings, and the Conclusion summarizes contributions, reflects on the findings, and suggests directions for future work.

1.2 Scope

This study focuses primarily on evaluating the system's performance rather than conducting in-depth user experience studies. While interaction plays a key role in the system's design, the research does not aim to analyze how users perceive or engage with the interface. Instead, the evaluation centers on the algorithm's behavior, with user behavior simulated for the purpose of conducting a quantitative assessment.

Additionally, while the proposed approach is designed to be generalizable to different domains, all experiments in this thesis are conducted within the context of audio-based search tasks. As a result, its applicability to other fields is not explicitly tested.

¹<https://github.com/aememis/IEMA>

Chapter 2

Background

This section outlines the fundamental concepts, methodologies and theoretical frameworks central to the thesis, along with a review of the relevant work from the literature, and the preliminary work.

2.1 Evolutionary Computation

Evolutionary Computation (EC) is a subfield of Artificial Intelligence focusing on solving optimization problems using principles inspired by natural evolution. EC relies on the concept of evolving solutions based on their fitness to the given problem. This approach is widely used in areas where traditional methods may struggle with complex, high-dimensional, or poorly understood problem spaces (Back and Schwefel 1996). Its application areas include optimization, machine learning, and creative tasks, like design and art generation (Cicirello 2024).

In natural evolution, life progresses through successive generations, with each building on the previous one, driving adaptation to the environment and improving survival and reproduction chances (Eiben and Smith 2015). This cycle inspired researchers to create algorithms simulating natural evolution mechanisms (Yao 2007; Eiben and Smith 2015). In Evolutionary Computation (EC), the key abstraction is the *individual*, which evolves over generations like organisms in nature. Individuals are represented by genotypes (genetic encoding) and phenotypes (observable behavior or performance), with the genotype serving as the data representation of the solution and the phenotype as the actual outcome of the genotype's expression (Fogel 1995). The representation of data in EA is crucial for the algorithms' success, making the effective abstraction of these concepts essential (Foster 2001). The following paragraphs describe how these concepts are adapted in an EA.

Gene In EAs, a gene is a basic unit of information representing a specific feature of a solution, which can be evolved through genetic operations.

Chromosome A chromosome is a set of genes that together form a complete solution. It undergoes evolutionary operations like mutation and crossover to evolve and improve.

Individual An individual is the entity representing the expression of a chromosome in the context of the EA. It refers to a solution that is evaluated in the evolutionary process based on its performance.

Genotype The genotype represents the genetic encoding of an individual, consisting of a set of genes. It defines the internal structure or parameters of a solution, which are manipulated during the evolutionary process to explore potential solutions.

Phenotype The phenotype is the observable expression of the genotype in the real world. It reflects the actual outcome or performance resulting from the genetic encoding, such as the selected audio samples in this case. While the genotype is the blueprint, the phenotype is the tangible manifestation of that blueprint.

Progeny Progeny are offspring generated through genetic operations such as crossover and mutation. They inherit traits from parent individuals and are evaluated for further evolution.

Population The population consists of all the individuals in a given generation. It evolves over time through selection, mutation, and reproduction to find optimal solutions. In this thesis, the term population is used regardless of the data type of concern, whether it is audio waveforms, audio features (high-dimensional), or projected data points (low-dimensional). In the evolutionary flow, the population refers to the set of individuals after being selected from the corpus based on their proximity to the closest discrete audio samples.

The evolution in EC can be described using Lewontin's mappings (Collins 2003; Lewontin 1974), which capture the relationship between genotype and phenotype spaces, illustrated in Figure 2.1. In this model, genotype space represents an individual's genetic code, while phenotype space corresponds to the observable traits or behaviors resulting from the genotype.

To implement Lewontin's mapping algorithmically, several key steps, based on Collins (2003), must be defined. First, the genotype is represented within the problem domain, forming the basis for evolution, with an evaluation function assigning a fitness score based on the phenotype. Selection and reproduction operators, including crossover and mutation, refine solutions across generations. Crossover allows two parent solutions to exchange genetic information, combining traits for potentially better outcomes, with single-point crossover being a common approach.

In single-point crossover, a random point within the genome array is selected, and genetic material beyond this point is swapped between two individuals. Genetic material beyond this point is swapped between the two individuals. The crossover point is randomly determined for each pair of individuals, with p_i where $i = 0$ to $n - 1$ representing the probability of crossover occurring beyond the i -th element of the individual's genome. An example of two-parent single-point crossover using binary numbers, with the crossover after the fifth element, is as follows:

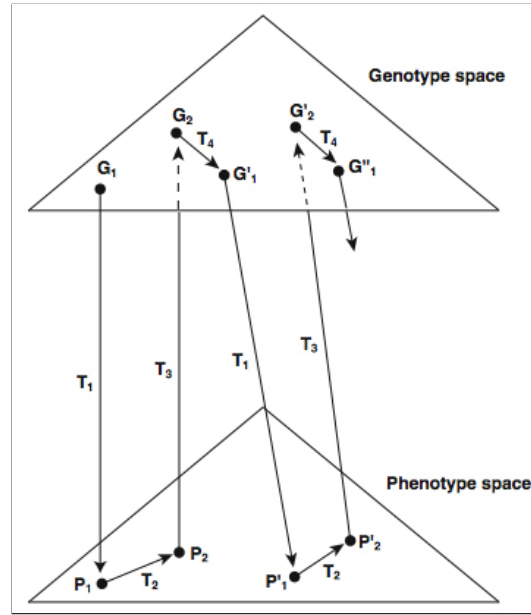


Figure 2.1: “Lewontin’s state space view of evolution. The caption reads: “Schematic representation of the paths of transformation of population genotype from one generation to the next. G and P are the spaces of genotypic and phenotypic description. G1, G’1, G2 and G’2 are genotypic descriptions at various points in time within successive generations. P1, P’1, P2 and P’2 are phenotypic descriptions. T1, T2, T3, and T4 are laws of transformation. ...” (Redrawn from Lewontin 1974)” (Wilkins, Stanyon and Musgrave 2012).

```

Parent 1:    0 1 0 1 1 | 1 0 1 0
Parent 2:    0 1 1 1 0 | 1 1 0 1
Offspring 1: 0 1 0 1 1 | 1 1 0 1
Offspring 2: 0 1 1 1 0 | 1 0 1 0

```

Mutation introduces random changes in an individual’s genetic code (Katoch, Chauhan and Kumar 2021). In Gaussian mutation, a gene value is adjusted by a random amount from a Gaussian distribution with a mean of zero and a defined standard deviation (Bell 2022). An example of Gaussian mutation applied to an individual in IEMA is as follows:

```

Original:    0 1 0 1 1 1 0 1 0
Mutated :    0 1 1 1 0 1 1 0 1

```

Lastly, proper combination of diverse methodological elements can lead to development of new interesting and more effective evolutionary methods (Kaliakatsos-Papakostas 2021). There are several methods that improve the performance and behavior of evolutionary algorithms. Quality Diversity algorithms (Pugh, Soros and Stanley 2016) aim to maintain a diverse population while optimizing for quality, which allows for a wider exploration of the solution space. Map-Elites (Mouret and Clune 2015) is one such approach that organizes the population in a grid, where each grid cell represents a distinct solution.

2.1.1 Interactive Evolutionary Algorithms

This concept builds on Dawkins' idea in *The Blind Watchmaker*, where natural selection is depicted as a *blind* optimization process, guided by the *watchmaker* of environmental pressures and survival challenges (Dawkins 1986). In IEA, human interaction adds an additional layer, shifting the algorithm from objective selection to one that includes subjective human evaluation.

IEA require a human user to actively participate in the selection process. This feedback guides the algorithm towards solutions that are more aligned with the user's preferences. The effectiveness of IEA lies in using human evaluative capacity, particularly in areas where subjective judgment is crucial or where defining an explicit fitness function is challenging. Moreover, IEA proved successful in making the evolution process user-centric and subjective. Hence, IEAs are found significant in interactive systems in fields like music composition, visual art, and design (Y. Wang and Pei 2024). Wide range of fields, including product development, industrial design, art creation, image retrieval, sound generation, and music composition, can be effectively addressed using IEA (Y. Wang and Pei 2024).

GenJam (Biles 1994) is one of the early examples of IEA in music, which used an interactive evolutionary system to evolve jazz solos. A user were included in the process to grade the elements in real-time (Biles 2013). More recently, Shao et al. (2010) implemented the Jive, a system incorporating IEA for musical composition. This system interactively modified a combination of piece-wise linear sequences to create melodic pieces of musical interest. They used physical controllers like a computer mouse or Wii remote for user interaction.

However, a significant challenge is the "fitness bottleneck," which is one of the main directions in the research of IEC (Pei and Takagi 2024). Fitness bottleneck defines the issue of requiring users to continuously evaluate and provide feedback on the evolving individuals. This manual evaluation process can be time-consuming and monotonous (Dahlstedt 2007). To address this, approaches like clustering the population have been proposed, where users only evaluate a representative sample from each cluster, reducing the number of evaluations required (Gong and Yuan 2011). Additionally, IEAs have been integrated with other technologies, such as neural networks and autoencoders, to reduce the fitness bottleneck and improve outcomes in applications like music composition (Masuda and Iba 2018; McDermott, Sherry and O'Reilly 2013).

2.1.2 Motion-based Interactivity in Evolutionary Systems

Motion-based interactivity introduces a novel, embodied dimension to human-computer interaction, particularly with IEAs (Wu and Zhao 2024). This approach has a potential particularly in creative domains where three-dimensional interaction comes real-life.

In the context of IEAs, motion-based interactivity significantly eases and enriches the feedback loop. Instead of relying on abstract control inputs (e.g., two-dimensional areas or sliders), users can simply move their bodies. This form of interaction also helps address common IEA limitations, such as user fatigue and ambiguous evaluation criteria (Pei and Takagi 2024).

Similarly in the preliminary work, as detailed in Section 2.8, running the system with a user in a lab setting was found to be a time-intensive process. Under similar configurations, a single evolutionary cycle (generating just one new generation) took up to 10 minutes, or in some cases, users might not complete a cycle at all within that time frame (Memis and Kelkar 2024). Given these constraints, conducting extensive tests with real users would be highly impractical.

On the other hand, to enable a quantitative approach to evaluating the system, the user’s interaction needs to be repeatable. Therefore, the inherently subjective nature of user interaction often needs to be set on a deterministic and repeatable basis. Instead of conducting tests with real users, a simulation was designed to mimic the user’s role in the system. Therefore, many works simulated the user behavior (Hagg, Asteroth and Bäck 2019).

2.1.3 Feedback Paradigms and Unary Feedback

In conventional EA, evolution is guided by predefined fitness functions that evaluate individuals based on a set of mathematical criteria. However, in IEA, this evaluation comes from subjective feedback. The way feedback is collected not only affects the user’s experience but also determines how selection pressure is applied and, consequently, how the system explores and refines its search space (Katoch, Chauhan and Kumar 2021).

IEAs employ various feedback mechanisms, ranging from explicit numerical scoring to comparative ranking or pairwise selection (Katoch, Chauhan and Kumar 2021). In IEMA, a unary feedback model is used, meaning users (real or simulated) can only provide positive reinforcement on the individuals. This paradigm mirrors interaction patterns found in modern digital platforms and social media, where engagement is often unary, either an item is liked and promoted, or ignored or discarded (Yang et al. 2024).

Unary Feedback is a commonly used paradigm in various fields where user engagement is simplified to a one-action decision-making, such as in recommendation systems, social media platforms, and content filtering (Pan et al. 2008; Yang et al. 2024). It has notable implications for the evolutionary process. Since there is no negative feedback, diversity must be maintained through other mechanisms rather than direct penalization. This shifts the role of exploration and exploitation, as selection pressure arises purely from accumulation of positive feedback rather than a balance of rewards and penalties (Z. Wang et al. 2021). While this approach simplifies user interaction and reduces cognitive load, it also presents challenges in ensuring a broad search without premature convergence (Jawaheer, Weller and Kostkova 2014).

2.2 Evaluating Evolutionary Algorithms

Evaluating Evolutionary Algorithms (EA) has been a key focus since the early days of evolutionary computation. Various approaches aim to test their performance using predefined test suites. One of the earliest and most consistent test sets was proposed by Jong and Alan (1975), which highlighted factors influencing algorithm performance and aimed to assess the general efficiency of EAs across different problems (Whitley et al. 1996).

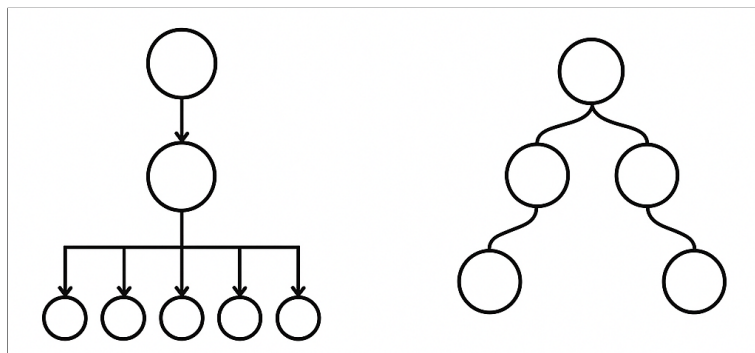


Figure 2.2: Diagram showing low (left) and high (right) phylogenetic diversity, where the left has a shallower evolutionary history, and the right a more diverse evolutionary history.

However, due to the user-centric nature of IEA, evaluating them brings different set of challenges. Often times there is no ideal set of solutions defined to comparatively analyze, which makes the most common analysis techniques less powerful. Additionally, the use of a discrete dataset in IEMA further complicates the definition of an optimal solution, as the search space does not allow for continuous exploration.

To evaluate IEA, researchers use alternative approaches. Performance is benchmarked through comparisons with existing algorithms, while qualitative evaluations focus on how well solutions meet user preferences. Quantitative evaluations often measure convergence rate, diversity, novelty, and the coverage of the explored parametric space (Shackelford and Corne 2004).

2.2.1 Diversity

Liebersson (1969) defined diversity as the range between homogeneity and heterogeneity in a population. In Evolutionary Algorithms (EA), diversity is crucial for global exploration and avoiding premature convergence (Drosou, H. Jagadish et al. 2017; Shir 2012; Squillero and Tonda 2016). Preserving diversity helps explore multiple optima, enhances crossover effectiveness, and increases the robustness of the algorithm, especially in dynamic problem landscapes (Sudholt 2020). Additionally, in interactive systems, diversity provides users with a broader range of solutions to explore and select from.

There are various approaches to assessing diversity in evolution. In ecological studies, Functional Diversity measures phenotypic characteristics, while Phylogenetic Diversity focuses on the evolutionary tree and the phylogenetic differences between species (Tucker et al. 2018). These methods highlight niche characteristics and the structural differences in evolutionary lineage respectively. Figure 2.2 illustrates an example of low and high phylogenetic diversity.

When comparing Functional and Phylogenetic Diversity measurements, Hernandez, Lalejini and Dolson (2022) showed that Phylogenetic Diversity offers distinct information from Phenotypic Diversity, as they found no consistent correlation between the two. Their findings make the two distinct diversity measures.

Finally, there are alternative approaches to diversity like the volume of the multidimensional convex hull, which capture the population’s spatial extent in

feature space (Cheng et al. 2024)

2.2.2 Novelty

Novelty-based diversity measures are another effective group of methods in estimating the population diversity. This measure defines the diversity based on the individuals seen in the past, and they aim for reducing the redundancy in a set of solutions.

Early approaches to diversity-based evaluation, like Maximal Marginal Relevance (MMR), focused on reducing redundancy while maintaining the relevance of retrieved documents and selected passages for summarization, thus balancing relevance with novelty (Carbonell and Goldstein 1998).

Later applications of novelty in Evolutionary Algorithms (EAs) include techniques like novelty search and novelty-driven fitness assignment, where the focus is on exploring new regions of the solution space. Novelty is crucial in evolutionary processes because it drives the system to explore uncharted areas and ensures that the search doesn't get stuck in local optima. This is especially important in complex search spaces, where the goal is not only to optimize a single objective but to foster creativity and diversity in the solutions (Lehman and Stanley 2011).

2.2.3 Coverage

Coverage-based metrics evaluate how well the evolutionary algorithm explores the solution space, either by assessing the coverage of the parameter space or the discrete parameters themselves. These metrics track the extent to which the solutions generated by the algorithm span the available space (Drosou, H. V. Jagadish et al. 2017).

It is commonly addressed quality in evaluating EA as it highlights the balance between exploration and exploitation in the evolutionary process. Coverage-based metrics provide a way to measure to what extent the algorithm has navigated through the search space, which is particularly important in problems with high-dimensional spaces (Pierrot et al. 2022).

2.3 Recommendation Algorithms in Music Systems

Recommendation systems in music have achieved significant success in recent years, particularly those designed to optimize music selection based on user preferences and behaviors. Such systems have become integral to music streaming platforms, improving user experience by leveraging algorithms that analyze audio samples and user preferences (Song, Dixon and Pearce 2012).

Advancements in music recommendation systems have explored a variety of approaches, each focusing on different aspects of user experience and data analysis, contributing to a broader understanding of how to deliver personalized audio recommendations. Some systems emphasize the use of audio features to enhance recommendations, while others prioritize user preferences or behaviors. Additionally, some systems aim to optimize navigation through large music collections (Liu et al. 2024).

MusicReco (Frantzvaag et al. 2025) explored a visual interface for presenting and interacting with a recommendation system, placing the user at the core of the design. They followed an iterative user-centered design approach and utilized visual interaction to enhance user engagement with the recommendation algorithm.

Alías, Socoró and Sevillano (2016) focused on enhancing recommendation accuracy by employing multiple audio feature extraction techniques. This approach overcame the limitations of traditional methods that often rely on textual attributes and improved the ability to deliver highly relevant recommendations. Additionally, Kostrzewa, Chrobak and Brzeski (2024) focused on the use of audio features for music recommendation, specifically examining the impact of different attributes. The authors developed a content-based recommendation system based on these features, and user testing demonstrated that MFCCs had the greatest influence on the representation and quality of recommendations.

Hybrid approaches have also been explored. Liu et al. (2024) combined genetic algorithms with recommender systems to gain a deeper understanding of user preferences, ultimately improving the recommendation process. Gao et al. (2025) integrated deep neural networks with a recommendation model to improve the recommendation process for multimedia content.

2.4 Audio Information Retrieval

Audio samples in their raw form are continuous waveform. While this representation is essential for playback, it is not directly useful for computational processing, especially in an evolutionary system where individuals must be represented based on their characteristics. Instead, extracting high-level audio features provides a more compact and relevant representation and captures essential aspects of the audio samples (Foote 1999). Audio feature extraction involves analyzing an audio signal to identify its key characteristics, and extracting relevant parameters that represent the essential traits of the signal (Alías, Socoró and Sevillano 2016). Extracting processable information from audio samples has various applications in fields like sound recognition (Ando 2001), music genre classification (George Tzanetakis and Perry Cook 2002) and source identification (Yuo, Hwang and H.-C. Wang 2005).

Extracting the right features from an audio signal is crucial for ensuring the success of audio processing systems. Selection and configuration of the extraction methodology highly depends on the task on hand (Alías, Socoró and Sevillano 2016). Evolutionary operations require individuals to be represented in a consistent format that allows for efficient manipulation. Each feature should contribute to defining an individual in the parameter space and ensure that variations between individuals reflect meaningful differences. The chosen features also determine the dimensionality of the parameter space. A balance must be struck between having enough features to represent variation and keeping the dimensionality manageable for computational efficiency.

In the light of these, there are several key aspects that guided the feature selection process in this work.

- **Energy and Loudness:** Features related to amplitude, such as Root Mean Square (RMS), help characterize the perceived loudness of a sample. This

is particularly relevant in distinguishing between percussive and sustained sounds (G. Tzanetakis and P. Cook 1999).

- **Timbre and Spectral Shape:** The harmonic content of a sound plays a significant role in individualizing an audio sample. Features like Spectral Centroid, Spectral Bandwidth, and Spectral Flatness describe how the energy is distributed across the frequency spectrum and help distinguish bright and tonal sounds from darker and noisier ones (Fu et al. 2011; Madhu 2009).
- **Temporal and Dynamic Changes:** Features like Spectral Flux capture how a sample changes over time, showing whether a sound is stable or has rapid fluctuations in frequency content (Kaminskas and Ricci 2012).
- **Perceptual Representation:** Mel-Frequency Cepstral Coefficients (MFCCs) are widely used in audio analysis as they approximate the way human hearing perceives sound. They help differentiate between different timbres and textures (Logan et al. 2000).

The extraction of these audio features is typically performed by analyzing the audio signal in smaller segments, often referred to as "windows," to capture variations in the signal over time. This allows for the extraction of features that are time-varying and represent the dynamic characteristics of the audio (Sharma, Umapathy and Krishnan 2020). There are various tools available for feature extraction, with librosa (McFee et al. 2015) being one of the most widely used for calculation of a wide range of audio features.

2.5 Audio Datasets

Audio datasets play a crucial role in the development and evaluation of systems that concerns sound. Numerous datasets have been created for various applications such as music genre classification, sound recognition, and environmental sound analysis. These datasets vary in size, complexity, and the types of audio they contain.

The corpus defines the possibilities within IEMA, but also imposes constraints. The system can only select, recombine, and mutate existing sounds, not generate new ones. Thus, the corpus acts as a boundary for the evolutionary process. To ensure meaningful exploration and evaluation, the dataset should be diverse, supporting gradual transitions and smoother variations. A limited dataset would restrict the evolutionary process and hinder the discovery of new solutions.

The dataset should also have a structured ontology that categorizes its samples systematically for the evaluation of system behavior in terms of category exploration and coverage. Additionally, a large enough dataset provides open-ended exploration without quickly exhausting available samples.

There are several open-source datasets that are suitable for this work. The ARAUS (Ooi et al. 2024) is an audio dataset of urban environments and includes a wide variety of environmental sounds. SONYC-UST-V2 dataset (Cartwright et al. 2020) is designed for urban sound classification and provides labeled environmental recordings. The UrbanSound dataset (Salamon, Jacoby and Bello 2014) contains sounds from urban settings, offering a rich collection of environmental sounds

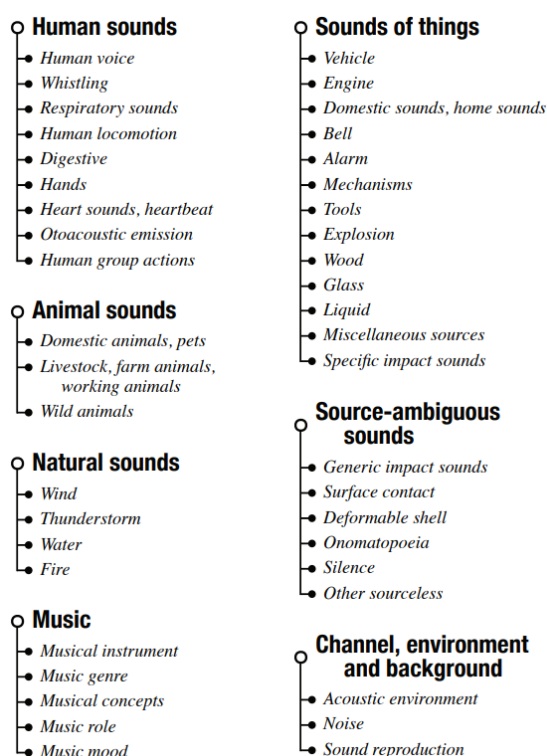


Figure 2.3: The top two layers of the Audio Set ontology (Gemmeke et al. 2017)

with various categories. Finally, FSD50K is an openly available dataset containing 51,197 human-labeled sound clips (Fonseca et al. 2022). These labels are 200 distinct categories based on the AudioSet Ontology (Gemmeke et al. 2017). It contains a total of over 100 hours of audio, with varying clip lengths ranging from 0.3 to 30 seconds.

The Audio Set ontology (Gemmeke et al. 2017) is a hierarchical classification system for audio events, organizing sounds into categories based on the audio events. It provides a multi-level taxonomy where broader categories branch into more specific subcategories. Figure 2.3 shows the top two layers in the Audio Set ontology.

2.6 Human-Computer Interaction in Music

Human-Computer Interaction (HCI) is the design and implementation of systems that facilitate communication between humans and computers in the context of user tasks and experiences. O'Brien, Rogers and Fisk (2008) highlights the importance of interactions that use prior experience and feed-forward methods to achieve functional and abstract goals. Advancements in motion capture, psychophysics, and machine learning have enabled more complex interaction mechanisms (Barbancho et al. 2013). Additionally, Music Interaction, as defined by Holland et al. (2013), refers to the design, evaluation, and use of interactive computer systems in music, drawing from HCI but influenced by the artistic and expressive constraints of music itself. This field has many branches and applications, with Interactive Music Systems being one prominent example.

2.6.1 Interactive and Adaptive Music Systems

Interactive Music Systems (IMS), as defined by Rowe (1992), are computer-based systems that change their behavior in response to musical input. IMS are used in dynamic interaction in both live performances and improvisational music contexts.

IMS have been a prominent field of research for decades, with extensive studies exploring various ways to integrate computer-based systems into musical performance. As highlighted by Perry Cook (2017), the development of new musical interfaces has been driven by advances in computing hardware, sensor technologies, and real-time sound synthesis. These systems aim to enhance the expressiveness and adaptability of musical instruments through intuitive control.

In the context of interactive music systems, Waisvisz's *The Hands* 1985 stands out as a significant example. *The Hands* integrated a set of controllers that respond to the performer's gestures. It allowed for dynamic, real-time control over the sound and facilitated an immersive experience. This makes it one of the pioneering systems in interactive music with the potential of gesture-based control in live performances and adaptive music contexts (Torre, Andersen and Baldé 2016).

Recently, with the rapid advancements in Artificial Intelligence, computing, and sensor technologies, IMS have reached new heights in creating immersive and adaptive musical experiences (L. Wang et al. 2024). These developments have enabled more sophisticated and seamless integration of virtual instruments, dynamic soundscapes, and AI-driven composition.

Moreover, adaptivity is one of the core elements in many applications of music information retrieval. It can help in application areas with high diversity, where some decisions have to be taken during runtime (Stober and Nürnberger 2013).

The AI-terity (Tahiroğlu, Kastemaa and Koli 2020) is a compelling example of adaptive music systems. It's an AI-based interactive instrument that offers a simple interface for adapting sound based on user preferences, and generates novel and unpredictable sound output.

2.6.2 Embodiment and Spatial Interaction in Music Systems

Embodiment, a term explored in philosophy, psychology, and engineering, is defined in various ways depending on the context (Kim and Seifert 2006). Kim and Seifert (2006) discuss embodiment in music systems as the integration of the physical body with algorithmic sound generation, where the body and virtual body interrelate without the physical body disappearing. Winkler (1995) views the body and space as musical instruments, highlighting their limitations due to physical factors like body part characteristics and space constraints.

Numerous interactive systems incorporate embodiment and spatial interaction. A notable example is *OtoKin*, a collaborative performance space where body movement control sound parameters (Dahlstedt and Skånberg 2019). The system maps body motion to specific sound controls, assigning different roles to body parts and participants within a three-dimensional performance space. Figure 2.4 illustrates these zones.

Additionally, Holland (1993) implemented an interactive system where they used a two-dimensional space as a control interface to generate harmonic music. They used the two-dimensional space as an interface for manipulating basic

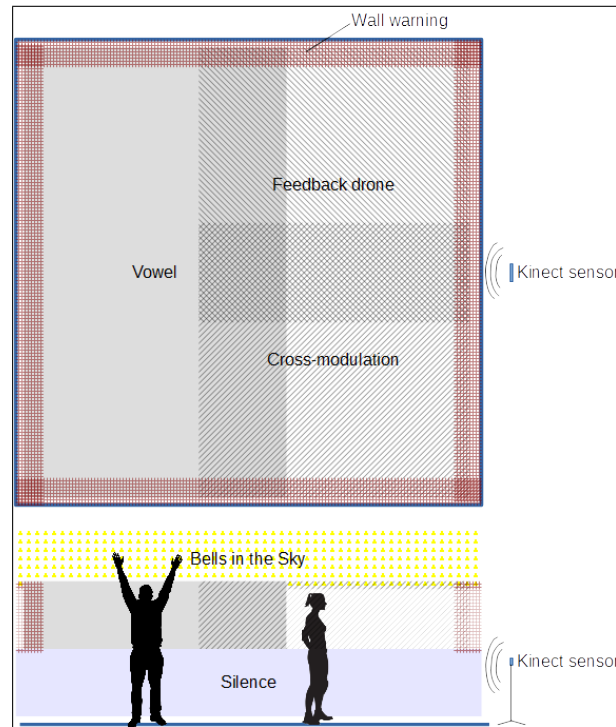


Figure 2.4: Map showing the zones for the sound engines, viewed from above and from the side. ... (Dahlstedt and Skånberg 2019)

harmony elements directly using a spatial interaction. This approach made certain tasks more tractable when compared to conventional methods.

There are many ways to enable spatial interaction, and one prominent approach is the use of infra-red motion capture technologies. It is widely used in many fields where tracking the body motion is essential, such as gaming, virtual reality, dance and interactive music systems (Kilpatrick III 2020). The motion capture technologies rely on reflective marker placement on the body to track movements in three dimensions.

Marker placement is crucial for effective motion capture integration. The head serves as a stable reference for spatial mapping, unlike other body parts, which may introduce complexity due to degrees of freedom and fatigue. Head-centric control enhances proprioceptive and vestibular alignment, making spatial interaction more intuitive (Vu and Proctor 2011; McGurk and MacDonald 1976).

Head movements are key in spatial hearing and sound source localization, with the brain using inter-aural differences and head-related transfer functions (HRTF) to determine sound positioning (Wenzel et al. 1993; Blauert 1997). These insights influenced the design of motion capture in IEMA.

2.6.3 Dimensionality Reduction for Interaction

High-dimensional data, common in fields like image processing and audio analysis, often contains intricate patterns but is difficult to manage due to its complexity and computational cost. This obstacle, commonly referred to as "curse of dimensionality", hinders processing power, reduces accuracy, and makes data difficult for humans to interpret (Jia et al. 2022; Donoho 2000). Dimensionality

reduction addresses this by mapping high-dimensional data to a lower-dimensional subspace, assuming data points lie on a low-dimensional manifold embedded in high-dimensional space, known as the *Manifold Hypothesis* (Cayton 2008; Fefferman, Mitter and Narayanan 2016).

Although dimensionality reduction may result in some loss of information, it is widely used in computational and data science for making complex data more accessible and interpretable. It reduces computational workload and maps data into a form easier for human observation and interaction, aiding exploratory data analysis (Ghojogh et al. 2023). In interactive systems, dimensionality reduction facilitates efficient user interaction with complex data by creating a projection layer that allows for dynamic, two-way binding between the user’s input and the data’s representation (Cavallo and Demiralp 2018).

Data can be mapped to lower-dimensional subspaces using linear or non-linear methods, broadly classified into spectral, probabilistic, and neural network-based approaches (Ghojogh 2021). Spectral methods use eigenvalue decomposition, probabilistic methods project data while preserving its distribution, and neural network-based methods learn a mapping from high-dimensional to lower-dimensional spaces.

PCA (Principal Component Analysis) (Hotelling 1933) is a linear technique that reduces data dimensionality by projecting it onto orthogonal components that capture the most variance. It is commonly used for tasks like noise reduction, feature extraction, and data compression to simplify analysis and interaction.

t-SNE (t-Distributed Stochastic Neighbor Embedding) (Van der Maaten and Hinton 2008) is a non-linear dimensionality reduction method used to visualize high-dimensional data. It preserves local structures by minimizing the divergence between high-dimensional and low-dimensional probability distributions.

2.7 Conceptual Spaces for Representation and Analysis

Data and information can be abstracted in different ways depending on the context of representation or analysis. The same underlying data can take on different forms depending on what aspects are being emphasized. For instance, in biological studies, an organism can be represented by its genetic information, physical traits, or evolutionary position, each offering a different perspective. Similarly, this thesis examines musical data across different domains, each highlighting specific aspects of the system, offering distinct ways to interpret how individuals evolve, interact, and are structured.

In many analytical approaches, data is represented in multiple domains to highlight different aspects. For example, in linguistics, a word can be analyzed phonetically as a waveform, syntactically in sentence structure, or semantically in a network of meanings. Moreover, in social network analysis, an individual can be represented by their interactions, role in the community, or demographic data. Similarly in EC, an individual can be examined through its parameter space, evolutionary lineage, or categorical identity, providing distinct yet interconnected insights.

This thesis uses three conceptual spaces to represent and analyze the data, each offering a unique perspective. These spaces will also form the basis of the

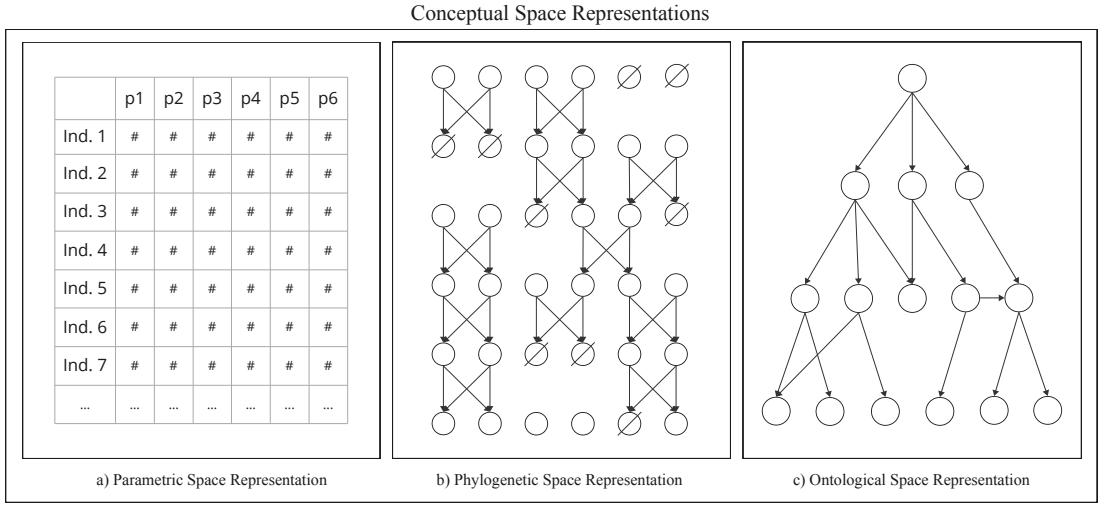


Figure 2.5: Illustration of the three conceptual spaces used in this research: (a) **Parametric Space:** Represents individuals as numerical vectors in a six-dimensional space, with parameters (p1–p6) as columns and individuals (Ind. 1–...) as rows. (b) **Phylogenetic Space:** Represents individuals’ evolutionary lineage in a directed acyclic graph, where nodes represent individuals, crossed nodes indicate unselected individuals, and edges show parent-child relationships. (c) **Ontological Space:** Represents individuals based on categorical classifications in a directed acyclic graph, where nodes represent categories and edges define hierarchical relationships.

evaluation strategy, as explained in Section 3.3. High-level data representations within these spaces are illustrated in Figure 2.5.

Parametric Space The Parametric Space, also commonly referred to as Behavioral Space, represents the space of parameters. It refers to the representation of individuals based on extracted audio features. Each individual in this space is defined as a vector of numerical values corresponding to its feature set. This space is continuous and serves as the foundation for measuring parametric assets in the data and performing numerical operations.

Phylogenetic Space Phylogenetic Space models the lineage of individuals over generations, representing their historical relationships in a tree-like structure. Each node represents an individual, and branches denote generational links. This space helps analyze how individuals evolve over time, their interrelations, and how the system explores the search space (Corts and Alachiotis 2023).

In this research, Phylogenetic Space captures the lineage of individuals across generations in a tree-like structure, where nodes represent individuals and branches denote generational links. This space allows analysis of how individuals evolve over time and their relationships in the evolutionary history.

Ontological Space Ontological Space represents entities based on categorical labels within a hierarchical structure. Many datasets use ontology-driven approaches to classify data, enabling structured analysis of similarities and differences (Gruber 1993). These categories form a tree structure, allowing for analysis of relationships, such as semantic distance between categories. This

approach is used in applications like recommendation systems and taxonomies (Xiao et al. 2018). In this thesis, Ontological Space describes individuals according to their categorical labels in a structured dataset ontology, with categories represented as nodes in a directed acyclic graph for analysis.

2.8 Related Work

Fornari (2016) presented an evolutionary algorithm to generate artificial soundscapes of birdsongs. Their system created a dynamic network of birdcalls through genetic operators emulating the sonic properties of a bird’s syrinx. This approach, aimed at mimicking natural soundscapes, did not aim for a single optimal solution but instead focused on continuously generating novel birdsongs. A key element of their system’s design was the real-time interaction where external data influence the evolutionary process.

Moreover, Jónsson, Hoover and Risi (2015) explored an interactive evolutionary computation (IEC) framework for sound synthesis, where users evolve novel timbres through the evolution of compositional pattern producing networks, and Neuro-evolution of Augmenting Topologies. Their approach allowed users to evolve new and unique sounds by manipulating a compositional network that can produce a variety of oscillatory waveforms. Their focus was on creating entirely new waveforms using a combination of mathematical functions, which could offer musicians greater flexibility and creativity.

Their interactive evolutionary framework aligns with the aim of this thesis, using user interaction to drive the evolution of audio samples. Moreover, the use of CPPNs and NEAT in sound synthesis echoes the use of genetic algorithms in IEMA, where the user’s feedback influences the evolutionary process.

Building on the previous work, Jónsson, Erdem et al. (2024) further explored the use of evolutionary algorithms for sound synthesis. They combined Compositional Pattern-Producing Networks (CPPNs) with Digital Signal Processing (DSP) graphs to generate novel and diverse audio outputs. The research uses Quality Diversity (QD) algorithms to guide sound exploration. The system also incorporates a deep learning model for categorizing and guiding the exploration. This work is another step toward creating an interactive environment where sound creation is both creative and technologically guided. This paper’s approach is closely related to this thesis, especially as it also combines evolutionary algorithms with novel exploration in the sound domain. The connection lies in how both projects use evolutionary strategies to explore large sound spaces and seek sonic novelty.

This thesis builds on the foundations established in our earlier work (Memis and Kelkar 2024), which applied Interactive Evolutionary Algorithms (IEAs) to body-motion controlled music interaction. In the IEMI system, we integrated Evolutionary Computation (EC) with motion capture, for real-time interaction with both continuous sound parameters and discrete data sets. Our research compared these approaches through quantitative tests. Additionally, we explored unary feedback, where users provided only positive feedback (e.g., button press to "like") in the IEA process. This work served as the primary motivation for this thesis, as detailed in Chapter 1.

The IEMI system involved two key prototypes. The first prototype focused on optimizing continuous audio effect parameters applied to real-time vocal effects and audio samples. The second shifted to a discrete parametric space, where evolutionary parameters selected the closest audio samples from a *soundbank* using Euclidean distance. Both prototypes were tested through user studies.

A significant finding from the IEMI work was the participants' positive response to the discrete audio sample selection mechanism, which provided more intuitive and perceivable interaction compared to evolving abstract parameters. Additionally, the unary feedback model proved to be effective and interesting in guiding the evolutionary progression without overwhelming users. These findings directly influenced this thesis, specifically the focus on discrete audio data, unary feedback, and nearest-neighbor selection strategy. Furthermore, the challenges in evaluating such systems in the preliminary work led to the development of a more systematic approach to evaluation metrics, as presented in this thesis.

Chapter 3

Methodology

This chapter outlines the methodology used to develop and evaluate the proposed system to address the research questions.

The system is designed to enable real-time interactive evolution within a spatial environment. It combines three key components, which are run in parallel: an Interactive Genetic Algorithm (IGA) with additional operations, a large dataset of audio samples, motion capture technology for spatial interaction, and a speaker array connected through an audio interface for playback. IEMA is designed to operate in a physical space equipped with motion capture and speakers.

IEMA primarily runs on a single computer, referred to as the "main computer" throughout the document. This computer is equipped with an Intel Core i7-9750H CPU running at approximately 2.60 GHz and 8 GB of RAM, and it operates on Windows 11 operating system. The algorithmic components of IEMA are implemented in Python, and details about its dependencies, installation, and usage can be found in the open-source IEMA repository¹.

The main computer runs multiple tasks in parallel threads. These include the Interactive Genetic Algorithm (IGA), dimensionality reduction and nearest neighbor selection; feedback acquisition; motion capture data handling; and playback. A summary of these tasks and their descriptions is provided in Table 3.1.

Thread	Task Description
Thread 1	IGA, Dimensionality Reduction & Nearest Neighbor Selection
Thread 2	User Feedback Listening & Processing
Thread 3	Motion Capture Data Listening & Processing
Thread 4	Audio Sample Playback

Table 3.1: Tasks run in parallel on the main computer.

A separate computer, referred to as the "mocap computer," is used for handling motion capture data, to which the mocap system is integrated. The role of this computer is further detailed in Section 3.2.1.

Figure 3.1 shows the technical diagram in the interactive room where the IEMA system operates.

Although IEMA is intended to work with real-time human input, actual users were not involved during testing. Instead, a simulated user model was

¹<https://github.com/aememis/IEMA>

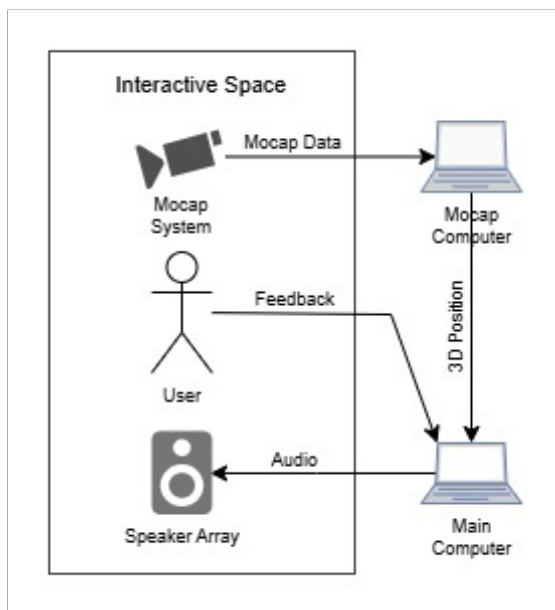


Figure 3.1: Diagram showing the connection and the data flow between mocap system, computers, network, and audio output components.

employed to generate consistent and repeatable feedback, ensuring controlled testing conditions. Section 3.3.1 details this approach.

The system’s performance was evaluated using a set of metrics designed to assess its functionality and inform further development. The following sections provide detailed descriptions of the system components and the evaluation methodology used in this study.

3.1 Algorithm

The main IEMA algorithm orchestrates various components, primarily the IGA, along with additional operations such as processing mocap data streams, extracting audio features, and performing nearest neighbor selection. This section provides an overview of the algorithmic flow in IEMA, summarizing how these modules work together, followed by a more detailed explanation of each individual module.

The algorithm runs in a cyclic manner and begins by initializing a population that is randomly sampled from the corpus. This population is then projected into a three-dimensional space using dimensionality reduction techniques, making it suitable for spatial interaction. The data is scaled to fit within the interactive space to allow the user to navigate within the environment and provide feedback.

During the interaction, the user navigates the room, with their position tracked in real time via a motion capture data stream. As the user listens to the audio samples and experiences the soundscape, they provide feedback in real time. This feedback is recorded and associated with the corresponding individuals. The process of feedback acquisition is detailed in Section 3.1.2. Once the selection is completed, the algorithm proceeds with crossover and mutation operations to generate new candidate individuals. A detailed explanation of the Interactive Genetic Algorithm (IGA) can be found in Section 3.1.2.

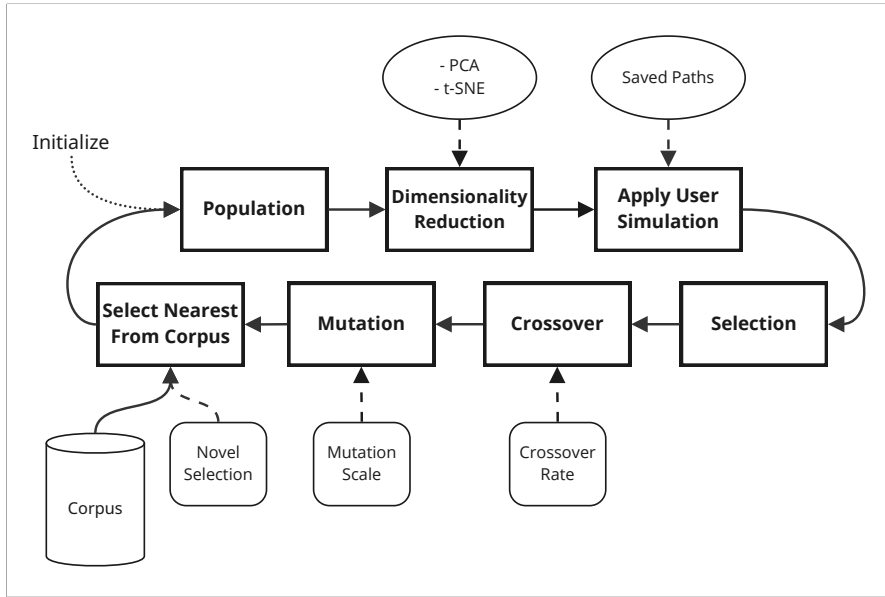


Figure 3.2: Overview of the IEMA algorithm workflow. The system begins with a randomly sampled initial population from the corpus. Each generation follows a pipeline involving dimensionality reduction, scaling, and simulated interaction through user path application. Selected individuals are then passed through crossover and mutation processes. Mutated individuals are matched to the nearest corpus samples to form the next generation.

However, since the search operates within a discrete space, the newly generated individuals do not directly correspond to existing samples. This intermediate population will be referred to as the "candidate population" throughout the thesis. At this stage, the algorithm retrieves the closest matching samples from the corpus, based on the Euclidean distance between the high-dimensional audio feature representations of the individuals and those of the samples. For each candidate individual, it is replaced by the closest match from the corpus.

This retrieval of the closest matches finalizes the generation of a new population. At this stage, data from each cycle, including selected individuals, discarded ones, and their evolutionary relationships, is saved to disk for later analysis. The updated population then undergoes another round of dimensionality reduction, and the process repeats. This flow is illustrated in Figure 3.2.

During this process, the algorithm assigns unique identifiers to each sample and tracks individuals across various representational spaces, including the reduced-dimensional interaction space, the evolving population, the dataset corpus, the audio waveforms for playback, and their positions within the phylogenetic history. This *two-way binding* provides interconnectedness among the individuals' representations. Details of this process are discussed in more depth in Section 3.1.7.

The following sections provide a detailed explanation of each part of the algorithm and its stage-wise prototypes.

3.1.1 Algorithmic Prototypes

The algorithm was developed through two main prototypes, iteratively refined based on evaluation outcomes: Prototype 1 (P1) and Prototype 2 (P2). While

both prototypes share the same core functionality, they differ in genetic algorithm implementation, testing parameters, and the scope of evaluation metrics. Each stage includes both an algorithmic component (A1 and A2), and a corresponding evaluation phase (E1 and E2). This prototype-driven development is depicted in Figure 1.2. This section details the algorithmic prototypes, A1 and A2, while the evaluation methodologies, E1 and E2, are discussed in Section 3.3.

The first prototype, referred to as A1, serves as the initial implementation of the IEMA system. It included the core evolutionary loop with basic selection, crossover, and mutation mechanisms, and used a simple pipeline for sampling from the audio corpus. Crossover was applied only to the selected individuals without any augmentation, and the remaining part was filled by randomly sampling from the corpus, regardless of whether the selected sample is being reused or not. This version was evaluated by the Initial Evaluation (E1), as explained in Section 3.3.2.

The transition from A1 to A2 was driven by key observations and design needs. In A1, crossover was applied only to selected individuals, leading to weak evolutionary pressure. To address this, A2 introduced augmentation, duplicating selected individuals for more extensive crossover, and a "novel selection" approach to prevent repeated sampling. Additionally, A1's rigid structure lacked flexibility for adaptive behavior, which A2 addressed by adding mutation scale control and allowing for PCA or t-SNE projection methods. This version was evaluated in three conceptual spaces: parametric, phylogenetic, and ontological, as detailed in Section 2.7. Tables 3.2 and 3.3 outline different design choices of the prototypes.

	Prototype 1 (A1)	Prototype 2 (A2)
Reproduction Approach	Selected Individuals Only	Augmented Amount
New Generation Formation	Repeated Selection Allowed	Novel Selection (No Repeated Samples)

Table 3.2: Selected algorithmic components in Prototypes 1 and 2

	Prototype 1 (A1)	Prototype 2 (A2)
Dimensionality Reduction Method	PCA Only	PCA or t-SNE
Genetic Parameters	Population Size, Crossover Rate	Population Size, Crossover Rate & Mutation Scale

Table 3.3: Controlled evaluation parameters in Prototypes 1 and 2

Algorithm 1 provides an example pseudocode for the IEMA. This pseudocode assumes a population size of 150, with 50 samples selected per iteration. PCA is used as the dimensionality reduction method, and crossover is applied to the selected individuals using a sliding window of size 1, which doubles the population size at that point. The remaining population is filled by sampling from the corpus after crossover. While these design choices are systematically varied during testing, the core functionality of the IEMA, as shown in the pseudocode, remains consistent across both prototypes, which is given in more details in the following sections.

Algorithm 1 Example IEMA Algorithm

```

1: Initialize:
2:   corpus  $\leftarrow$  extractAudioFeatures(dataset)
3:   populationSize  $\leftarrow$  150
4:   numberOfIterations  $\leftarrow$  100
5:   newGeneration  $\leftarrow$  selectRandomPopulation(corpus, populationSize)
6:   generations  $\leftarrow$  [newGeneration]
7:   path_bank  $\leftarrow$  loadSavedPaths()
8:
9:   for each path  $\in$  path_bank do
10:    for iter  $\leftarrow$  1 to numberOfIterations do
11:      proj3D  $\leftarrow$  applyPCA(newGeneration)
12:      proj3D  $\leftarrow$  normalize(projection3D)
13:      selectedIndividuals  $\leftarrow$  applySelection(newGeneration, proj3D, path)
14:      recombined  $\leftarrow$  applyCrossover(selectedIndividuals)
15:      mutated  $\leftarrow$  applyMutation(recombined)
16:      closestFromCorpus  $\leftarrow$  getClosestSamples(mutated, corpus)
17:      previousGeneration  $\leftarrow$  newGeneration
18:      randomFromCorpus  $\leftarrow$  sampleFromCorpus(corpus)
19:      newGeneration  $\leftarrow$  closestFromCorpus + randomFromCorpus
20:      evolutionaryTree  $\leftarrow$  updateTree(newGeneration, previousGeneration)
21:      generations.append(newGeneration)
22:    end for
23:  end for

```

3.1.2 Interactive Genetic Algorithm

The proposed algorithm is population-based and is an interactive genetic algorithm where the fitness criteria are defined solely by external input. The initial population is randomly sampled from the corpus of audio tracks, where each track characterized by a set of extracted audio features. Each individual represents 6 audio features normalized and scaled between 0 and 1, as explained in section 3.1.4. Therefore the initial population has the shape of (N, 6), where N is the selected population size.

The algorithm proceeds with selection based on the user's feedback. Following this, crossover and mutation are applied in sequence to generate the candidate population. Detailed descriptions of each evolutionary operation are provided in the following sections.

Feedback & Single-like Paradigm

The way feedback is collected not only affects the user's experience but also determines how selection pressure is applied and, consequently, how the IEMA

explores its search space. In the IEMA, as the user interaction is simulated, only the latter aspect is considered.

IEMA is designed around a "unary feedback" mechanism where users can only provide positive reinforcement by selecting individuals they prefer. This means that rather than a continuous or graded evaluation, evolution is driven solely by affirmative selections, similar to the "like" interactions commonly seen in social media interfaces.

The feedback is received through a physical button that the user had during the interaction, and the algorithm continuously monitors these clicks during the evolutionary process. Upon receiving a click, the system assigned a fitness score of 1 to the corresponding individual. Details of this feedback acquisition is given in Section 3.2.

In the case of simulated interaction, described in Section 3.3.1, button clicks are programmatically triggered at specific positions along the predefined paths. Each simulated click follows the same process of identifying the nearest individual and assigning the binary fitness values accordingly.

All feedback events, including the selected individuals' identifiers, generation numbers, and corresponding fitness values, are recorded and stored in tabular form. This data is later used for evaluation, as explained in Section 3.3.

Selection

The selection process involves identifying and retaining individuals that have been positively rated by the user. In IEMA, the selection occurs once a predefined threshold of user ratings is achieved. This threshold is a hard-coded value that determines when enough feedback has been received to trigger the selection, thus proceeding with the creation of a new generation. The selected individuals are then passed to further genetic operations, and the unselected ones are deleted.

The selection threshold is set to 50. This value was chosen to provide a manageable number of samples for real-time interactive selection, while making sure that a meaningful subset of the population is evaluated. Given the population sizes of 100, 200, and 350 for testing, as discussed in Section 3.3.4, selecting 50 individuals represents a substantial portion of the population. This allows for effective evolutionary pressure while still leaving space for the introduction of new samples, without overloading the user.

Recombination

The recombination phase involves the exchange of genetic material between selected individuals to create offspring, thereby generating new generations. Following the selection, the population undergoes recombination through a two-parent crossover. At this stage, genetic material of the two selected individuals is exchanged to create two new offspring. This exchange is performed using single-point crossover strategy, which is explained in Section 2.1

The probability of crossover occurring (crossover rate) is a predefined parameter which determines the likelihood of crossover for any given pair within the population. This parameters is systematically varied during the testing, as explained in Section 3.3.4.

When only a subset of the population is selected for crossover, evolutionary pressure may be too weak, which can limit IEMA. To address this, the number of individuals involved in recombination is increased by duplicating selected individuals, expanding the crossover pool. This augmentation is controlled by the augmentation ratio. This parameter is set to either 0 (disabled) or 0.5 during testing. When set to 0.5, half of the selected individuals are randomly added to the crossover pool. This allows each individual to appear in multiple pairings across the recombination process. The testing strategy is explained in Section 3.3.4. The individuals are randomly paired for crossover operation, and the resulting offspring are then passed to the mutation stage. An example of crossover applied to a pair of individuals in IEMA is as follows, where the blue and red portions represent the crossed-over segments:

```

Parent 1      :    0.13 0.76 0.54 | 0.06 0.73 1.34
Parent 2      :    0.34 0.21 0.85 | 0.45 0.63 0.06
Offspring 1   :    0.13 0.76 0.54 | 0.45 0.63 0.06
Offspring 2   :    0.34 0.21 0.85 | 0.06 0.73 1.34

```

Mutation

Mutation occurs after crossover to introduce genetic variation into the population. This is achieved using a Gaussian mutation strategy, as detailed in Section 2.1.

Each gene in an individual's genome has a predefined probability of mutation, determined by the "mutation rate." This rate, set to 0.05, controls the likelihood of each gene being selected for mutation in each generation and remains constant across all tests.

When mutation occurs, the offspring generated in the previous recombination phase are iterated through, and each gene is selected for mutation based on the predefined mutation rate. If selected, the gene is altered by a deviation drawn from a Gaussian distribution with a mean of zero. The magnitude of the mutation is controlled by the "mutation scale" parameter, which dictates the standard deviation of the distribution. This parameter is controlled during the testing phase, with details provided in Section 3.3.4.

An example mutation applied to an individual in IEMA is as follows:

```

Original : 0.13 0.76 0.54 0.06 0.73 1.34
Mutated  : 0.13 0.69 0.43 0.06 0.85 1.34

```

Figure 3.3 illustrates the evolutionary process and key terms involved in a sample evolutionary history within the IEMA system.

3.1.3 Dataset

The dataset, also referred as corpus, forms the search space for IEMA. Each individual in the population corresponds to one of these samples from the corpus, and evolution involves selecting, modifying, and replacing samples from the dataset. Therefore, no new samples are generated, evolution simply explores and reshapes the existing corpus.

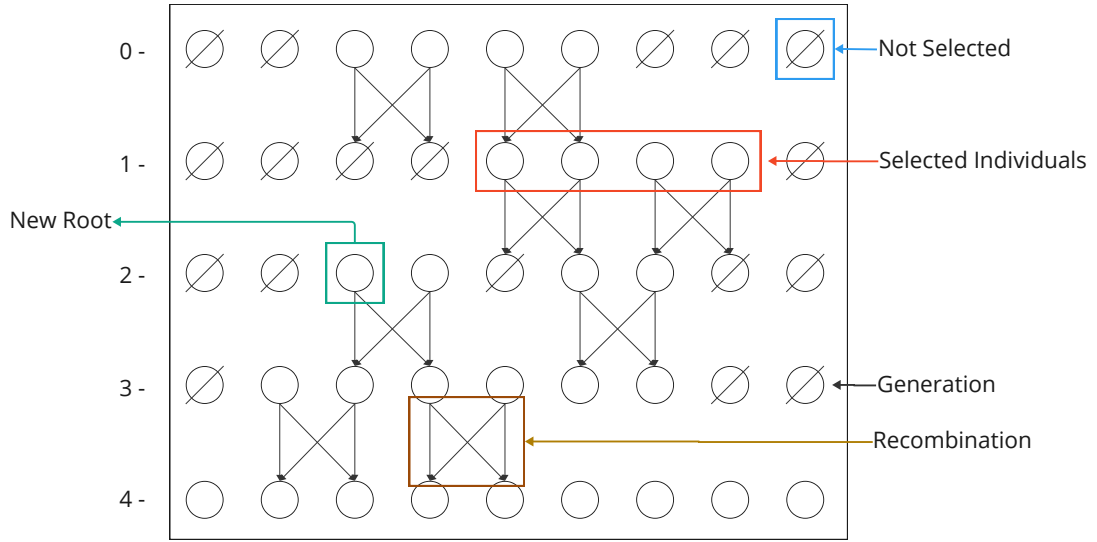


Figure 3.3: Diagram illustrating the history of an example evolutionary process in Interactive Evolutionary Computation (IEC). The diagram highlights key stages of the evolutionary process with corresponding annotations, including the selection and non-selection of individuals, the root node, the progression of generations, and the recombination process

For this work, the FSD50K dataset was selected due to its diverse audio samples and well-structured categorization. Its broad range of categories and durations makes it well-suited for that IEMA, ensuring that the corpus provides rich diversity and a good basis for systematic evaluation. More details about the FSD50K dataset and its qualities that informed its selection for this thesis can be found in Section 2.5.

The dataset was preprocessed for further refinement and structuring. The preprocessing included sample selection based on duration, and transforming the dataset’s ontology for structured analysis.

Data Preprocessing

Given the scale of the dataset and the computational requirements of the evolutionary system, the development set of FSD50K, which contains 40,966 samples, was determined to be a sufficient subset for this work.

Audio samples of varying lengths are beneficial as there is no strict constraint on uniform duration, unlike tasks that require fixed-length pooling or segmentation. The interactive nature of the IEMA allows for a more natural representation of audio, where different durations can coexist without disrupting the evolutionary process. For reference, Figure 3.4 shows the histogram of clip lengths in the dev set of the FSD50K dataset.

Extremely short or excessively long samples could introduce issues. Samples shorter than 0.5 seconds were removed to avoid overly brief clips that might not contain noticeable sonic characteristics. Similarly, samples longer than 20 seconds were excluded to maintain consistency in interaction and avoid excessively lengthy sounds that could dominate the exploration process during the user interaction. This filtering resulted in 35976 samples.

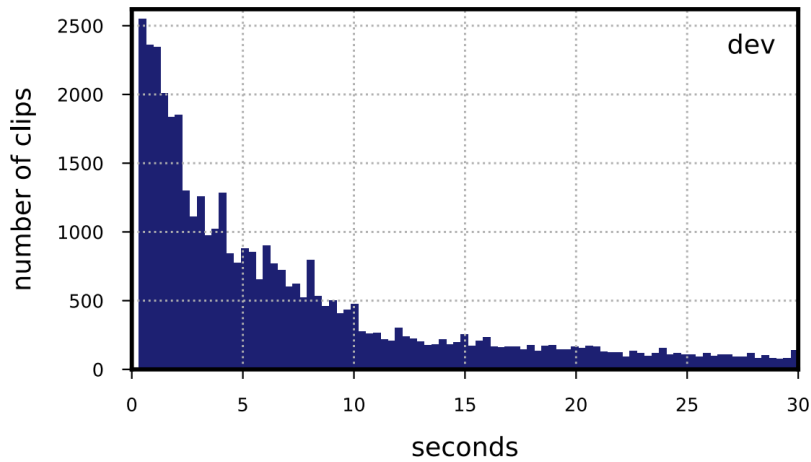


Figure 3.4: Audio clip length distributions in dev ... (Fonseca et al. 2022)

Preprocessing of the Dataset Ontology

The Audio Set ontology originally released in a nested dictionary format, where categories were structured as parent-child relationships. This format makes direct comparisons between categories difficult, as the hierarchy is not optimized for the analysis conducted in this thesis, such as calculating the ontological distances between the categories.

To address this, the ontology was transformed into a directed acyclic graph, where each category was assigned a level based on its appearance in the graph. The graph edges pointed from the parent category to the child categories. The root category was assigned level 0, its direct children were assigned level 1, and so on. Since some categories appeared in multiple parent branches, the earliest occurrence was used as the reference level for consistency. By assigning levels, it became possible to evaluate categories at the same depth in the hierarchy. This restructuring did not modify the data, but provided a standardized representation for comparisons. It also allowed for a more optimized and consistent category-based analysis across different samples. Figure 3.5 shows a plot of the directed acyclic graph representation of the Audio Set ontology. This visualization is intended to provide an overall structural overview of the ontology, rather than detailed information about individual elements.

For storing and manipulating the directed acyclic graph representation of the dataset ontology, the `networkx` is used (Hagberg, Schult and Swart 2008), a Python package for analysis of network-based data structures and algorithms.

3.1.4 Feature Extraction

The feature extraction process aimed to get representative and discriminative information from the raw audio data. For this purpose, a set of audio features was used to capture various acoustic properties of the audio samples. Since the IEEMA operates in a discrete search space, where each individual corresponds to a unique audio sample, these features serve as the primary data representation for evolutionary operations.

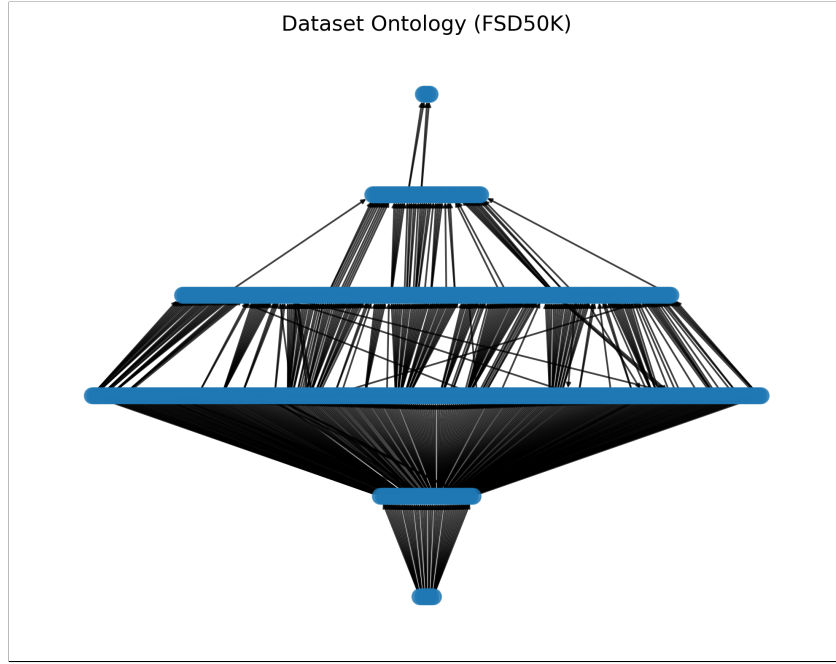


Figure 3.5: A plot of the Audio Set ontology represented as a directed acyclic graph (DAG). The blue circles denote the categories, with relations indicated by arrows pointing from the child nodes to the root. The topmost super root node, is created to establish a single root data structure. This figure provides a general visual overview of the ontology and its hierarchical structure. The graph is plotted with the root positioned at the bottom.

Six specific features are selected for their ability to capture qualities like energy, loudness, timbre, spectral shape and perceptual characteristics, as explained in Section 2.4. The selected features are given in Table 3.4.

The selected measures are calculated for each audio sample using sliding windows. The window size is set to 1024 samples, with a step size of 512 samples between consecutive windows. For each window, the relevant feature values are computed. The resultant feature array is averaged across the audio signal to get the mean value. This process results a dataset of audio features representing the audio corpus, with six descriptive numbers per audio sample. The audio features are extracted using *librosa*, a Python package for music and audio analysis, which is discussed in Section 2.4. To ensure consistency, the extracted features are normalized by scaling each feature within the range of 0 to 1 based on its minimum and maximum values.

3.1.5 Nearest Neighbor Selection

In IEMA, the evolutionary process operates continuously, but the underlying dataset consists of discrete audio samples. This means that when a new individual is generated through reproduction, its exact representation does not necessarily exist in the dataset. Hence, the system approximates the intended individual by selecting the closest existing sample from the corpus based on its numerical audio features. To achieve this, IEMA applies nearest neighbor selection using Euclidean distance. The generated individual's numerical feature representation is compared to all available samples in the dataset, and the closest matching sample

Table 3.4: Selected Audio Features

Selected Feature	Description
Root Mean Square (RMS)	Measures the energy or loudness of the signal. Useful for distinguishing between quiet and loud sounds.
Spectral Bandwidth (SB)	Represents the spread of frequencies around the spectral centroid. Indicates the complexity or richness of the sound.
Spectral Flux (SFlux)	Measures the rate of change in the power spectrum. Useful for detecting transient sounds and differentiating between sustained and percussive elements.
Mel-Frequency Cepstral Coefficients (MFCC)	Captures the spectral shape of a sound by approximating human auditory perception. Useful for distinguishing between different sound textures.
Spectral Centroid (SC)	Indicates the perceived center of mass of the spectrum. Useful for differentiating between bright and dark sounds.
Spectral Flatness (SFlat)	Measures how noise-like or tonal a signal is. Useful for distinguishing between harmonic and noisy sounds.

is selected as the representative of that individual, and replaces it. This enables to function in a discrete search space while maintaining evolutionary operations as if they were continuous. On the other hand, this approach introduces an inherent approximation error, as the chosen sample will not perfectly match the intended parameters of the newly generated individual.

During evaluation phase, two versions of the nearest neighbor selection strategy were compared. The first allowed reselection, where individuals were chosen from the corpus solely based on proximity, regardless of whether they had been selected in previous generations. The second enforced novelty, where if the closest match had already been selected before, the algorithm continued to the next closest, and so on, until an unused sample was found. The comparative testing approach is explained in Section 3.3.3.

3.1.6 Dimensionality Reduction

To make the high-dimensional population of audio samples available for spatial interaction, it is projected into a 3-dimensional space using dimensionality reduction techniques. These techniques enable the adaptation of the high-dimensional data to fit within the physical confines of a 3-dimensional space e.g., a room.

Principal Component Analysis (PCA) and t-distributed Stochastic Neighbor Embedding (t-SNE) are used as the dimensionality reduction methods. These were selected not only for their efficiency and popularity, but also to compare during the testing phase, as they are implemented in distinct ways and they preserve different aspects of the data's structure. Since interaction in IEMA is with the

projected data, the choice of projection algorithm plays an important role on how individuals are spatially arranged, and thus how interaction unfolds. PCA and t-SNE, one being linear, and other being stochastic and non-linear respectively, offer distinct perspectives. These dimensionality reduction strategies are discussed more in detail in Section 2.6.3, and the how they are comparatively tested is detailed in Section 3.3.3.

3.1.7 Two-way Binding

As briefly introduced earlier in the chapter, dimensionality reduction would map the data points to new ones represented in another space. Hence, the points lose their representation in the higher-dimensional space. The IEMA is designed to have both lower dimensional space and higher dimensional space control each other, referred to as two-way binding. This means a data point that is being interacted with in the lower dimensional space is directly tied with the original data point in the higher dimensional space. This would make the lower dimensional space an interface for actually interacting with the original higher dimensional space in real-time.

To uphold the associations between the data and its projection, the system is designed to track these points by unique identifiers, e.g., indexes. The two-way binding mechanism is implemented by indexing the data points in both the higher-dimensional and lower-dimensional space. To reduce the computational load, look-up tables are created to store the indexes of data points in the high-dimensional space alongside their corresponding points in the three-dimensional space. This allowed associating feedback on a certain data point in the interactive space, to the corresponding data point in the audio features data. This binding mechanism also included the corresponding audio waveforms for each individual for real-time playback.

3.2 User Interaction

IEMA is ultimately intended for real-time user interaction, which takes place within a physical lab space.² The user interaction is facilitated by the motion capture system, the feedback button, and the speakers. The motion capture system provides a continuous stream of three-dimensional location data, which is combined with the projected point cloud in real-time to make the user navigating through the population. The feedback button facilitated the user feedback acquisition. Finally, the spatial audio system provided the user with the sonic output of the IEMA.

Although IEMA is intended for real users, to ensure the comparability and reproducibility of the experimental results, the evaluation of the system are based on simulated user interaction rather than actual human. In that case, an artificial user interacts with the system. The rationale behind this approach, along with its use in similar works, is discussed in Section 2.1.3, while the implementation details are given in Section 3.3.1.

²Throughout the document, regardless of whether it involves a real user or a user simulation, the environment that the user is navigating is referred to as the “interactive space.”

3.2.1 Motion Capture

The physical place for the user interaction was the "MCT Portal," a laboratory for network music communication and interactive music systems located in the University of Oslo ³. It is a rectangular room equipped with an infra-red (IR) motion capture system with 8 cameras positioned around the ceiling. The motion capture (mocap) component of the system is facilitated by this mocap system.

Although implemented with a specific mocap system, IEMA is designed to be adaptable to any environment equipped with a real-time tracking setup. The system does not depend on a specific brand or model of mocap hardware. It requires only a representative control point that can be continuously streamed during interaction.

A single reflective marker placed and secured on top of the user's head. This placement is chosen for accurate and intuitive interaction, the rationale of which is explained in Section 2.6.2.

The network connection between mocap computer and the main computer is established over a local area network (LAN). The data is streamed as UDP packages with a frequency of 120 Hz. The packages consisted of data points, each referring to one instance of three-dimensional user position recorded at a time. The network packages are received by the main computer, and are integrated into the IEMA algorithm, details of which are given in Section 3.1.

3.2.2 Feedback Acquisition

The feedback is received through a physical button that the user had during the interaction, and the algorithm continuously monitors these clicks during the evolutionary process. For ease of use, a Bluetooth mouse was used as the physical button for providing feedback during the interaction. Upon receiving a click, the system calculates the Euclidean distance between all individuals in the current population and the user's position in the parametric space. The individual with the smallest distance, which is also the current individual on playback, is identified as the target of the feedback. This individual is assigned a fitness score of 1.

3.2.3 Auditory Feedback

Auditory playback is a key component of IEMA that facilitates real-time interaction as users explore sound samples. The system generates audio through the Python module, as detailed in section 3.1. As users navigate through samples, multiple audio files may play simultaneously, which is managed by an audio buffer capable of handling up to 12 parallel samples.

The audio is routed through the MIDAS M32 audio interface ⁴, connected to the main computer, and played through a 12-speaker array mounted on the ceiling, as shown in Figure 3.6. Although spatial audio is not central to IEMA, the existing real-time tracking and speaker array could support spatial audio control if needed.

³<https://www.hf.uio.no/imv/english/about/rooms-and-equipment/mct-portal/>

⁴<https://www.midasconsoles.com/product.html?modelCode=0603-ADX>

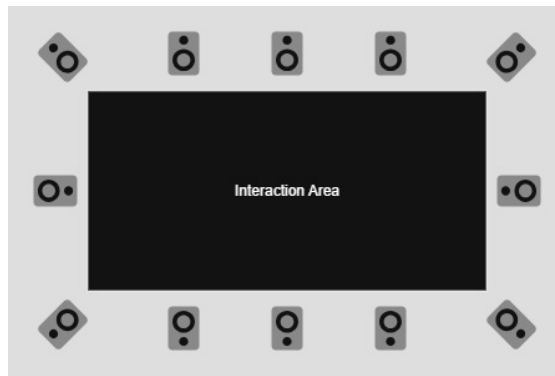


Figure 3.6: Diagram showing the speaker array placement in the lab room where the real user interaction happens.

3.3 Evaluation

This section outlines the methodology used to evaluate the IEMA, including the design and implementation of the evaluation process, formulation of evaluation metrics, and how they are applied.

3.3.1 User Simulation

To create a simulation that mimics the user's role in the system, the interaction process needed to be abstracted into its core components. In IEMA, the key aspects of user interaction are navigation and feedback. Therefore, the user simulation is designed to replicate these functionalities by modeling a random user moving through the interaction space and providing positive feedback at predefined locations.

User's paths within the interactive space are pre-generated and used for testing. This ensures that the system operates as if a real user were actively engaging with it while maintaining repeatability for systematic evaluation. These paths consist of randomly selected points, each representing a location where the simulated user provides feedback, i.e., "liking", on the audio sample at that point. The length of the paths are determined by the selection threshold of 50 individuals per generation, as explained in Section 3.1.2.

The paths are generated using a script that creates random points within the boundaries of the interactive space, sequentially. The generated paths are saved as sequence of numbers representing the three-dimensional position, and can be reused consistently across multiple test runs.

3.3.2 Initial Evaluation (E1)

The purpose of the initial evaluation (E1) was to gain a basic understanding of how the IEMA system behaves under different parameter settings. Specifically, it focused on analyzing how the algorithm explores the parametric space throughout the evolutionary process. For this evaluation, data recorded during each generation of evolution was used. E1 applied metrics to measure how the population of individuals varied over time. The primary metrics used in this stage are Functional Variance (FV) and Dataset Coverage (DC).

Functional Variance

Functional Variance (FV) is commonly used in evolutionary algorithms to assess the diversity of individuals within a population. In the context of this system, it measures the variance of individuals in the parametric space, based on their extracted audio features.

Tracking FV under different system configurations aims to provide an initial understanding of the effect of the configuration parameters in IEMA, and also an understanding on how the system behaves, specifically whether the system maintains variation or converges prematurely in the parametric space.

This metric specifically captures the spread of feature values in the parametric space by computing the variance of individuals' audio features within the population at each generation. The mathematical formulation of FV is:

$$FV(P_t) = \frac{1}{|P_t|} \sum_{x \in P_t} (x - \bar{x}_t)^2$$

where:

- P_t is the population at generation t
- $|P_t|$ is the number of individuals in P_t .
- x represents the feature vector associated with an individual.
- $\bar{x}_t$ is the mean feature vector of all individuals in P_t

A high FV indicates that the population is diverse, with individuals spread across a wide range of feature values in the parametric space. Conversely, a low FV indicates that the population has converged, with individuals clustering around a narrow range of feature values.

Dataset Coverage

Dataset Coverage (DC) tracks the total number of unique data samples from the dataset that have been introduced into the population during the entire evolutionary process.

DC is formulated as:

$$DC = \frac{|\bigcup_{t=0}^T P_t|}{N_{\text{dataset}}}$$

where:

- P_t is the set of individuals in generation t .
- $\bigcup$ represents the union of all unique individuals introduced into the population over all generations
- N_{dataset} is the total number of unique samples in the dataset.

A high value indicates that the system has sampled from a wide range of dataset instances, meaning that the evolution process is reaching different regions of the dataset. A low value suggests that many dataset instances remain untouched, which may indicate an overly narrow search space or selection pressure favoring only certain samples.

The Initial Evaluation (E1) provided insights into how the algorithm's parameters affected the IEMA's behavior of converging over the course of an evolutionary process. It helped understanding how the parameters affect data convergence/diversity of the IEMA. It also provided insights into what more is needed to be able to evaluate the IEMA more representatively. Finally, this analysis was insufficient to fully understand the system's behavior, as it lacked critical aspects introduced by the unique design choices of IEMA. The rationale of improving from E1 to E2 are further detailed in the next section (3.3.3).

3.3.3 Final Evaluation (E2)

This section describes the methodology behind the Final Evaluation (E2), which aims to analyze the IEMA system through a more detailed and structured approach. Building on the insights from the initial prototype (P1), E2 expands the evaluation scope by applying a broader set of metrics that target specific system configurations and behaviors. The E2 is designed to be applied on the second implementation (A2).

The limited set of metrics in E1 provided only a partial view of how the system behaved. While useful for basic observation, these metrics did not capture key aspects of IEMA's operation, such as how evolutionary paths developed, how categories shifted over time, or how broadly the Audio Set categories were explored. To address this, the second evaluation phase (E2) introduced a more comprehensive set of metrics across multiple conceptual spaces. This included phylogenetic metrics to trace evolutionary history, categorical metrics to measure diversity and change in ontological terms, and coverage metrics to assess how much of the dataset categories were actually reached.

This expanded metric set aims for a more complete and nuanced understanding of the IEMA's internal behavior. The evaluation itself (E2) is designed not to identify an optimal version of IEMA, but to understand how different system configurations behave under varied settings, and how these behaviors align with different interaction goals. The IEMA system is not defined by a fixed function or solution, instead, it is a flexible, user-driven system, and the evaluation reflects that flexibility.

The evaluation proceeds in several stages. First, crossover rate and mutation scale parameters are tested systematically to observe their effects on the system's ability to generate diversity and novelty. This large-scale parameter sweep provides the basis for selecting a smaller set of representative configurations.

To demonstrate how the IEMA system can be configured for different interaction goals, three desired scenarios were defined. Each scenario reflects a practical use case where certain behaviors of the system are prioritized regarding diversity and novelty.

Scenario 1: Diversity-Oriented Configuration In some use cases, the goal may be to build an interactive system where the user is presented with a broad variety of audio samples, covering different sonic characteristics, sources, or categories. For example, a designer may want the system to offer a mixture of contrasting sounds, such as percussive, vocal, and ambient samples, making sure that the population within each generation contains samples that are distinct from one another. In this scenario, the focus is on maintaining diversity throughout the evolutionary process, though introducing entirely new samples each generation is not necessarily required.

Scenario 2: Novelty-Oriented Configuration Another possible use case prioritizes the continuous introduction of new audio material throughout the evolutionary process. In this scenario, the goal is to ensure that the system consistently explores new data by regularly refreshing the population with audio samples that differ as much as possible from those previously encountered. This configuration can be suited for creative contexts where the emphasis is on exploration and unpredictability, and where the diversity within each individual generation is less important than the ongoing discovery of novel content.

Scenario 3: Combined Diversity and Novelty Configuration In some cases, it may be desirable to design a system that emphasizes both diversity and novelty simultaneously. For instance, a designer may want the system to present a spread of varied audio samples within each generation while also ensuring that new and different samples are introduced over time. This could be useful in contexts where maintaining both a wide sonic palette and a continuous sense of discovery are equally important, such as in dynamic sound installations or exploratory music tools.

Moreover, these selected configurations are further examined by introducing additional parameters. These parameters are the choice of dimensionality reduction algorithm, and the activation of a novelty selection mechanism. These extended tests aim to understand how different design choices affect the IEMA's behavior towards different use cases. The results of these tests are given in Results & Discussion chapter.

Throughout this process, the same set of evaluation metrics is used across different configurations to provide consistent analysis. These metrics are not used to judge the system in terms of success or failure, but as tools to describe, interpret, and compare how different configurations behave. The metrics operate across three conceptual spaces, each offering a different angle on diversity, novelty, and the structure of the evolutionary process.

This staged approach allows for both exploration and explanation: by progressively refining the system and observing its behavior through a fixed set of analytical tools, E2 shows how a flexible, user-driven evolutionary system like IEMA can be built, adjusted, and evaluated in relation to different goals.

The E2 uses a range of metrics. Some of these metrics are straightforward statistical measures, while others combine multiple factors to capture more complex relationships in the evolutionary process. Each metric is designed to capture a specific perspective on the system's behavior, ensuring that they do not overlap in what they measure.

Functional Space These metrics operate within the functional (phenotypical) space, focusing on how the numerical representation of individuals evolves over time. These metrics analyze the statistical properties of the population, such as variance, dispersion, or clustering within the feature space, to assess how the system navigates the search space.

Phylogenetic Space These metrics focus on evolutionary history, tracking how individuals evolve over generations and their relationships to one another by analyzing their placement within the evolutionary tree structure.

Metrics in phylogenetic space are designed to evaluate the extent to which the interactive evolutionary algorithm explores the solution space over generations. Unlike feature-based diversity metrics that examine each generation in isolation, these metric considers the entire evolutionary lineage and analyzes how solutions are derived and how they evolved over time. These metric is crucial because they evaluate long-term evolutionary behavior, rather than just analyzing within a single generation. It complements other metrics by offering a historical perspective on how evolution unfolds in the system.

By incorporating the metrics in the phylogenetic space, the evaluation methodology ensures that the system is assessed not just in terms of its momentary diversity but also in its long-term exploratory behavior and lineage evolution.

Evaluating the IEMA in the phylogenetic space requires answering two key questions:

1. Is the algorithm refining existing solutions or exploring new ones?
This assesses whether the population is primarily composed of descendants from long-standing lineages or whether it is frequently disrupted by new samples introduced from the corpus. A system that relies too much on a small set of early roots may indicate convergence to a narrow search region, whereas frequent disruptions by new samples may suggest a lack of continuity in evolution, potentially preventing adaptation.
2. How divergent are individuals from each other within the evolutionary tree?
This measures how much the population diverges over time, assessing whether offspring remain closely related to their ancestors or whether the system produces highly distinct variations. If divergence is low, it suggests the algorithm may be stagnating and producing similar solutions, while high divergence indicates continuous novelty and exploration.

To address these questions, two complementary metrics are proposed: Pairwise Distance (PD) and Root Contribution Index (RCI). Formulations of these metrics are detailed in Chapter 4.

Categorical Space Lastly, these metrics focus on the ontological space, which is based on the dataset's predefined category structure. These metrics analyze the distribution and transitions of semantic categories within the population, as defined by the dataset ontology. Instead of measuring numerical variation, they assess how diverse the explored categories are, how frequently new categories emerge, and whether the system revisits previously explored ones.

The metrics in the categorical space are designed to evaluate how the categories of audio samples evolve over generations. Unlike feature-based diversity metrics, which focus on variations in continuous feature space, these metrics specifically analyze discrete category changes as defined by the dataset ontology.

Evaluating the ontological space requires answering two key questions:

1. How diverse are the categories present in the current generation?
This examines whether the system explores a broad range of dataset categories or remains confined to a few.
2. How much do the categories drift from one generation to the next?
This assesses whether the system continues to explore new regions of the dataset or stabilizes around a set of categories.

To address these aspects, two metrics are proposed: Categorical Diversity (CD) and Category Change Rate (CCR), formulations of which are given in Chapter 4.

The metrics aim to assess the system from three main aspects: diversity, novelty, and coverage. Each aspect captures a different characteristic of the evolutionary process and provides insight into how the system explores and organizes solutions in the evolutionary process. Diversity examines how varied the individuals are within a population, while novelty looks at the introduction of new solutions over time. Finally coverage measures how much of the search space has been explored.

3.3.4 Testing

The proposed evaluation metrics are applied across different system configurations to analyze both the behavior of the evaluation framework and how the algorithm responds under varying conditions. The focus is not only on assessing the algorithm's performance but also on understanding how the designed metrics capture different aspects of the system's evolution. To achieve this, multiple configurations of key system parameters are tested. These configurations are systematically varied under the proposed evaluation metrics.

For each configuration, IEMA is executed multiple times, corresponding to the number of paths used in the cross-validation. To minimize biases and improve the reliability of the results, the testing process followed 5-fold cross-validation. Each predefined interaction path is tested 5 times to ensure that results are not skewed by a particular path's structure and that the evaluation remains robust.

These runs are pre-configured, and the program iterates through all configurations automatically. A *run* refers to a full evolutionary cycle of IEMA, lasting for a 100 generations under a specific configuration while following a predefined path. Since paths are reused across configurations, each configuration is executed once for each path. Throughout these runs, various data is logged in real-time for later analysis, as detailed in Section 3.3.4

System Parameters

To analyze the behavior of the system under different conditions, key parameters were selected to create multiple configurations of the IEMA. These parameters were chosen to represent two core aspects of the system: evolutionary mechanisms

and interaction/projection dynamics. From the evolutionary aspect, population size and crossover rate were selected as they directly influence how individuals evolve over generations. From the interaction and projection aspect, the choice of dimensionality reduction algorithm was included, as it affects how the high-dimensional data is structured for user interaction. By systematically varying these parameters, multiple configurations were tested to evaluate the IEMA.

Crossover Rate The crossover rate defines the proportion of selected individuals that undergo crossover to produce offspring in each generation. A higher crossover rate encourages more mixing of traits, potentially leading to faster exploration of the search space, while a lower rate maintains more continuity from previous generations. This parameter directly influences the rate of genetic recombination. Therefore, it plays a key role in balancing exploration and exploitation within the evolutionary process. For testing, the crossover rate was set to 0.5, 0.65, and 0.8.

Population Size The population size determines the number of individuals in each generation. It defines both the initial set of samples drawn from the corpus and the fixed size maintained throughout evolution. When the population decreases due to selection, it is replenished by introducing new samples from the corpus to maintain the defined size. For testing, the population size was set to 100, 200, 350.

Mutation Scale The mutation scale controls the magnitude of change applied to individuals during the mutation step. While mutation introduces variation by altering individual parameters, the scale determines how large these changes are. A low mutation scale results in minor adjustments, maintaining proximity to the original individuals, whereas a higher scale applies more significant shifts, which potentially move individuals into new regions of the search space. In IEMA, mutation operates in the parametric space before mapping to discrete audio samples, meaning the scale affects how far individuals may jump before being matched to the nearest sample. For testing, the mutation scale was set to 0.01, 0.025 and 0.05.

Dimensionality Reduction Method The choice of dimensionality reduction algorithm affects how individuals are placed and clustered, which in turn influences how they are explored and selected during evolution. Since the user (or simulation) interacts with the projected space rather than the full feature space, different methods may lead to different evolutionary outcomes. This makes it particularly interesting to examine how different algorithms impact the interaction dynamics and evolutionary behavior. To investigate this effect, three widely used dimensionality reduction techniques were tested: PCA, t-SNE, and UMAP, each with distinct properties in preserving global and local structures.

Novel Selection from Corpus The novel selection as a parameter controls whether previously selected audio samples can be reused, or if only unseen samples from the corpus are allowed to participate in the evolutionary process. When novel selection is enabled, it prevents repeated selection of the same sample from the

corpus. When disabled, repeated selections are allowed. This parameter was tested in enabled and disabled states.

Data Collection

Most of the data and events are saved in real-time, while some additional processing is performed afterward based on the collected data. Each run's data is systematically stored for later analysis in the evaluation phase.

The collected data spans across the three conceptual spaces. In the parametric space, original audio feature values of individuals and the corresponding indexes for the original audio samples are saved for each iteration. In the evolutionary space, parent-child relationships between individuals are tracked and stored as a directed acyclic graph. Finally, in the ontological space, categorical labels are retrieved from the Audio Set ontology for each individual, based on the audio samples that they represent. This ontological information is then used to calculate evaluation metrics using a graph representation of the Audio Set ontology, which is detailed in Section 3.1.3.

The aforementioned data is collected for each user simulation path applied during the evaluation process.

Final System Description chapter follows with the final prototype (P2). It details the final algorithmic configuration (A2) and the final evaluation metrics (E2). The outcomes are analyzed and discussed in Section 5.

Chapter 4

Final System Description

This chapter presents the final configuration of the IEMA algorithm (A2) and the evaluation metrics (E2). For A1, it highlights the key differences introduced in A2. For E2, the chapter explains the formulation, purpose, and how the metrics capture various aspects of the system's behavior. The details of the implementation and design choices from P1 to P2 are discussed in the previous chapter.

4.1 Algorithm Configuration

The final algorithmic prototype (A2) configuration retains all the core functionality as explained in Section 3.1, with several key enhancements. A2 introduces augmented crossover, where the number of individuals selected for crossover is increased by randomly duplicating a subset of selected individuals. Specifically, 50% of the selected individuals are randomly chosen for duplication, which expands the pool of candidates. This method increases evolutionary pressure by allowing for a broader range of individuals to recombine.

Additionally, A2 integrates the Novel Selection mechanism, which ensures that each individual is selected only once during the entire evolutionary process. When the algorithm selects individuals for the next generation, it checks the historical records of the population to confirm that previously selected individuals are not included again. This way the population continually explores new areas of the search space without revisiting prior solutions.

A2 also compares the t-distributed Stochastic Neighbor Embedding (t-SNE) for its success in handling complex data with non-linear techniques, distinguishing it from PCA's linear approach. Lastly, A2 introduces the mutation scale as a configurable parameter, which is further explained in Section 3.3.4.

4.2 Evaluation Metrics

This section details the formulation of the evaluation metrics. The variables and notation used in these formulations are summarized in Table 4.1 for reference.

4.2.1 Functional Variance

The Functional Variance (FV) metric is unchanged from the initial evaluation (E1), explained in section 3.3.2. It continued to keep the evaluation from the functional

Variable	Description	Type
P	Population	Set
$\mathcal{T}$	Evolutionary tree	Directed Acyclic Graph
x	Individual (associated data)	Vector
v	Node (vertex) representing an individual in the phylogenetic history	Node
d	Distance function between nodes or categories	Scalar
C	Set of categories associated with an individual i	Set
c	Category associated with an individual i	Label
$Desc$	Function for the number of descendants of a node	Scalar
Age	Function for the age of a root node	Scalar

Table 4.1: Description of variables, notations, and data types

diversity of the individuals within a generation, in the high-dimensional discrete parameter space.

4.2.2 Phylogenetic Space

Root Contribution Index

The Root Contribution Index (RCI) measures the influence of early vs. newly introduced solutions in shaping a population. It does this by calculating the contribution of each root node, representing the original individuals introduced at the start or through reseeded by counting their number of descendants and scaling it by the age of the root (in generations).

The formula for RCI is:

$$RCI(\mathcal{T}) = \frac{1}{|\mathcal{T}|} \sum_{v \in \mathcal{T}_{root}} \frac{Desc(v)}{age(v)}$$

where:

- $\mathcal{T}$ represents the evolutionary tree, which, despite being a directed acyclic graph (DAG) due to crossover and mutation mechanisms, is referred to as a tree to emphasize the hierarchical ancestral relationships.
- $|\mathcal{T}|$ is the total number of nodes in the tree.
- v is a root node, i.e., an individual that has no parents.
- $Desc(v)$ is the number of descendants of a root v .
- $age(v)$ is the number of generations since the introduction of a root v .

A high RCI indicates that older roots contribute heavily to the population, meaning the system is favoring refinement over disruption. This suggests that

the evolutionary process is producing long-standing, stable lineages, potentially improving solutions over time.

On the other hand, a low RCI suggests that newly introduced roots contribute more to the population, meaning the system frequently replaces previous evolutionary paths with new individuals from the corpus. This may indicate constant disruption, which can be good for exploration but may hinder convergence.

Pairwise Distance

The Pairwise Distance (PD) metric quantifies how divergent individuals are from each other within the evolutionary tree. Unlike RCI, which focuses on the origin of individuals, PD measures the spread of individuals in the phylogenetic space and aims to reveal whether the system maintains diverse evolutionary trajectories or clusters around similar solutions.

$$PD(P_t) = \frac{2}{|P_t|(|P_t| - 1)} \sum_{\substack{v_i, v_j \in P_t \\ i < j}} d(v_i, v_j)$$

where:

- P_t is the population at generation t
- $|P_t|$ is the number of individuals in P_t .
- v represents the data associated with an individual from P_t
- $d(v_i, v_j)$ is the phylogenetic distance between two individuals in the tree.

A high PD indicates that individuals are highly divergent, meaning that evolutionary trajectories are well-separated and promoting exploration.

On the other hand, a low PD suggests that individuals are closely related in the evolutionary tree, which means that the system is producing similar solutions rather than exploring new directions.

4.2.3 Categorical Space

Categorical Diversity

The Categorical Diversity (CD) metric measures the spread of unique categories present in a given generation, normalized by the population size. This provides insight into how much of the dataset's categorical space is actively being explored at any point in evolution. This normalization ensures that the metric is not biased by the population size, allowing for direct comparison across generations.

$$CD(P_t) = \frac{|set(C_{P_t})|}{|P_t|}$$

where:

- P_t represents the population at generation t
- C_{P_t} denotes the number of unique categories among individuals in P_t .

- $|P_t|$ shows the total number of individuals in P_t .

A high CD indicates that many different dataset categories are represented in the current population, suggesting broad exploration of the dataset. In contrast, a low CD means that the system is focusing on fewer categories, potentially indicating convergence or bias toward certain dataset regions.

Category Change Rate

The Category Change Rate (CCR) measures how much the categories shift between generations. It is calculated by measuring the categorical distance between an individual and its parents and averaging it across the population.

$$\text{CCR}(P_t) = \frac{1}{|P_t|} \sum_{i \in P_t} \left(\frac{1}{|\text{Cat}_i| \cdot |\text{Cat}_{\text{parents}}|} \sum_{c_i \in \text{Cat}_i} \sum_{c_p \in \text{Cat}_{\text{parents}}} d(c_i, c_p) \right)$$

where:

- P_t is the population at generation t
- Cat_i is the set of categories associated with individual i
- $\text{Cat}_{\text{parents}}$ is the combined set of categories across both parents.
- where d is the distance function between two categories within the dataset ontology.

A high CCR suggests that categories are changing significantly between generations, meaning the system continues to explore new dataset regions. In contrast, a low CCR suggests that the system is stabilizing, consistently selecting from similar categories across generations.

4.2.4 Coverage

Coverage metrics assess how much of the dataset has been explored by the algorithm over the course of an evolution. The coverage metrics measure the extent to which the system has accessed the dataset, both in terms of categories and individual data points.

Category Coverage

This metric calculates the total number of unique categories from the dataset that have appeared in at least one generation throughout the evolutionary process. It provides insight into whether the system is sampling broadly across all possible categories or if it remains constrained within a limited subset of dataset categories.

$$\text{Category Coverage} = \frac{|\bigcup_{t=0}^T C(P_t)|}{C_{\text{total}}}$$

where:

- P_t represents the population at generation t .
- $C(P_t)$ is the set of unique dataset categories represented in P_t .
- C_{total} is the total number of categories available in the dataset.
- $\mathcal{T}$ represents the evolutionary tree.

A high value indicates that the system has accessed a broad range of dataset categories, suggesting strong exploratory behavior. A low value suggests that many categories remain unvisited, meaning the system may be biased toward certain parts of the dataset or struggling to generalize.

Dataset Coverage

The Dataset Coverage (DC) metric is unchanged from the initial evaluation (E1), explained in section 3.3.2. It continued to assess the coverage of discrete data samples after an evolutionary process. Unlike category coverage, which looks at high-level structure, DC measures how many distinct items from the dataset have actually been explored.

Chapter 5

Results & Discussion

This chapter presents the results and discusses the findings in relation to the research questions.

5.1 Initial Prototype

This section presents the results of the initial evaluation using two metrics: Functional Variance (FV) and Dataset Coverage (DC), for different combinations of population size and crossover rate.

The results regarding the initial prototype provided a solid foundation for understanding how the system behaves and where improvements could be made. The results demonstrated that the core system functions as expected. Crossover contributed to convergence in the population, as seen in the declining functional variance. On the other hand, larger populations led to noticeably higher dataset coverage, confirming that increasing the number of individuals naturally supports broader exploration. This effect is mainly due to the fact that new individuals are sampled from the corpus after each reproduction to fill the population. These results validated the basic behavior of the system and confirmed that, with a minimal setup, IEMA responds to core evolutionary parameters as expected.

As shown in Figure 5.1, higher crossover rate slightly decreased the Functional Variance (FV). Population size also had an effect: variance at population size 200 was more stable than at population size 100. These results suggest that crossover had limited impact in driving exploration in this setup, likely because the number of selected individuals was small, and the system filled the rest of the population with random new samples from the corpus. This likely diluted the effect of recombination and made it hard for crossover to shape the population.

On the other hand, Dataset Coverage (DC) increased with population size. At population size 100, coverage stayed below 10%, while at 200 it rose above 30%. However, even in the higher population, coverage remained well under 40%, which suggests limited exploration of the dataset. Increasing the crossover rate from 0.5 to 0.8 resulted in lower coverage in both population settings. This aligns with the idea that stronger crossover reinforces recombination within the current population rather than introducing variety, leading to less new material being sampled from the dataset. These results are shown in Figure 5.2. The plots also include the *ideal maximum coverage*, calculated as the total number of individuals generated across all generations (population size $\times$ number of generations) divided by the

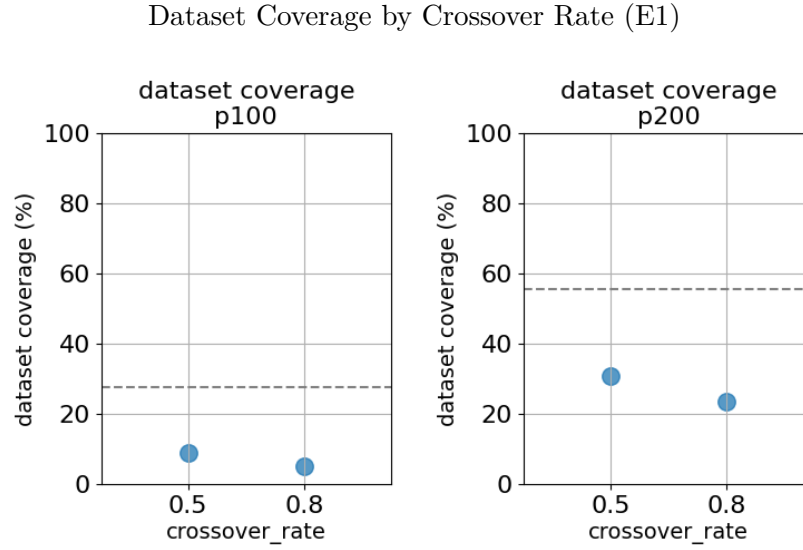


Figure 5.1: Dataset Coverage in percentage in the Initial Evaluation (E1) across varying crossover rates and population sizes. The dashed line indicates the *ideal maximum coverage* in 100 iterations for given population size.

total number of samples in the corpus. This value serves as a theoretical upper bound assuming no repeated selections during the process.

The results from E1 revealed two important limitations in the system’s behavior. First, the effect of crossover was very limited. Because, relative to the population sizes, only a small subset of the population was selected by the user and the number of crossover pairs remained low. This weakened the evolutionary pressure and made it difficult for the system to converge or meaningfully evolve over generations. Crossover alone could not drive significant changes in the population under these conditions.

Second, it became clear that many of the new individuals sampled from the corpus were often repeated. Since the system allowed for reselection of the same samples without constraint, this reduced the overall novelty of the population and the coverage remained low. These two observations pointed to the need for changes that would both increase the influence of crossover and improve the system’s exploratory behavior.

Another important aspect that remained outside the scope of this prototype was the role of the projection methods. Similarly, the set of tested parameters was limited to a few, which left room for broader investigation. Finally, mutation-related parameters were fixed throughout, meaning that its potential role in shaping the behavior of IEMA was not yet examined.

5.2 Final Prototype

This section presents the results of the final prototype. This stage evaluates final algorithm prototype (A2), which includes system improvements, additional parameters, and an extended set of metrics, which are detailed in Chapter 4. The section first presents the outcomes of the IEMA configuration phase, where

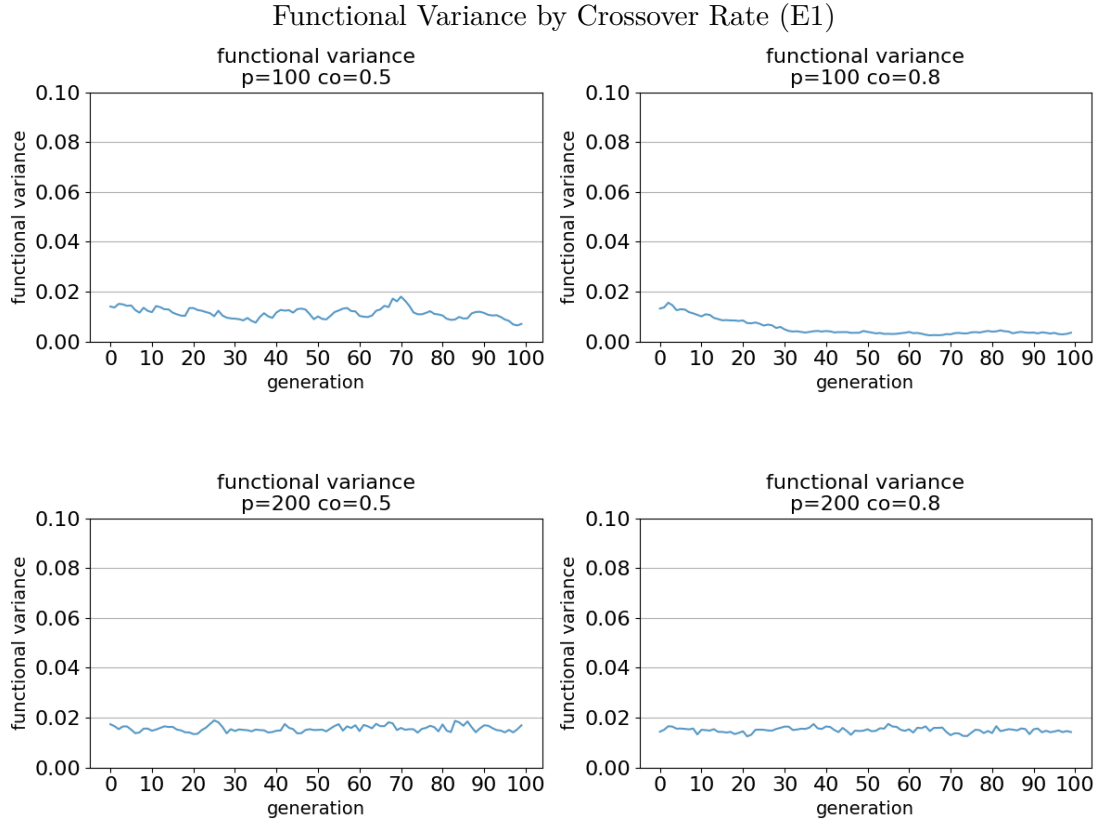


Figure 5.2: Functional Variance (FV) in the Initial Evaluation (E1) over generations on varying population sizes and crossover rates. Rows are population sizes, columns are crossover rates. x-axis is generation number and y-axis is functional variance value.

various parameter setups were tested to construct specific scenarios of the system behavior. These scenarios are explained in Section 3.3.3. It then shows the results of evaluating additional IEMA-specific parameters on the selected configurations, using the proposed metrics to both further inform system design and assess how effectively the final metrics capture IEMA's behavior.

Towards defining specific desired IEMA configurations, the following sections present the results of the proposed evaluation metrics applied across various configurations of IEMA, based on variations in crossover rate, mutation scale, and population size. These results are used to identify the configurations that align with the desired scenarios. The evaluation metrics introduced in the final prototype (E1) are given in Chapter 4, desired system behaviors (scenarios) are detailed in Section 3.3.3, and selection of system parameters is explained in Section 3.3.4.

5.2.1 Functional Variance

The Functional Variance (FV) showed a decreasing trend influenced by both population size and crossover rate. In general, higher population sizes tend to maintain variance for longer, while higher crossover rates accelerate its reduction. This effect is more pronounced when population size is smaller and crossover rate is higher. Conversely, lower crossover rates allow variance to persist over more generations and delay convergence.

These results show how the balance between crossover rate and population size shapes the exploration of the parametric space. Crossover works to homogenize the population by recombining individuals, but its impact depends on how many individuals are present. In smaller populations, crossover has a stronger effect in concentrating the population more quickly. In larger populations, however, its effect is diluted.

Figure 5.3 presents the functional variance against varying crossover rate and population size.

Functional Variance by Crossover Rate and Population Size

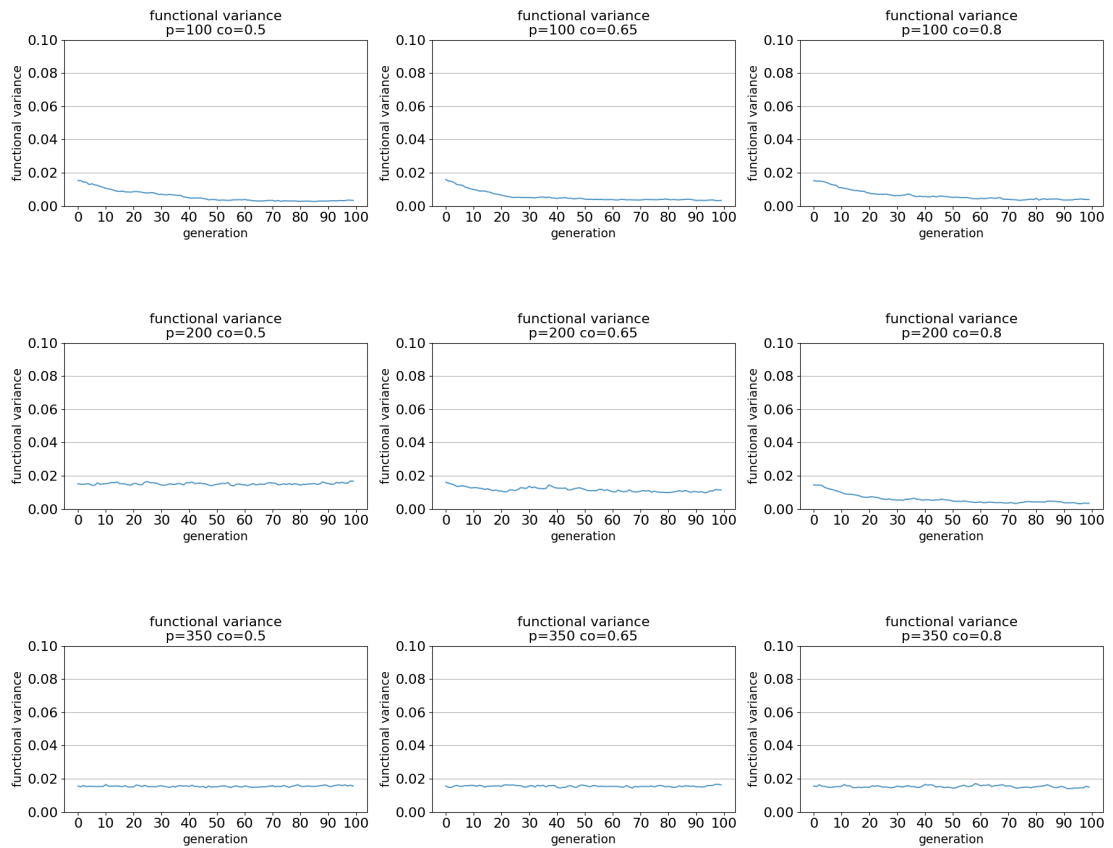


Figure 5.3: Functional Variance (FV) over generations on varying crossover rates and population sizes. Rows are varying population sizes, while columns are varying crossover rates. x-axis represents generation number, and y-axis represents functional variance value.

FV was also influenced by mutation scale and population size. Across all

population sizes, lower mutation scales (0.01) resulted in a faster drop in variance. As the mutation scale increased, IEMA was able to maintain slightly higher variance over generations. However, the effect remained relatively small, especially in larger populations.

While not as prominent as the effects seen with crossover, mutation scale still showed a measurable influence on Functional Variance within the IEMA system. This effect reflects how mutation in IEMA introduces controlled variation without entirely destabilizing the population. Since mutation operates on individuals after crossover in this system, its influence depends not just on rate but also on the structure of the population it acts on. In that sense, Functional Variance helps confirm that mutation in IEMA contributes only modestly to maintaining parametric spread, and that, if desired, diversity needs be reinforced through other means when population size is low.

Overall, the results show that, in IEMA, the given mutation scale range alone does not drastically affect functional variance, but when combined with larger population sizes, it helps maintain a more stable and diverse parametric spread over time.

Figure 5.4 presents the impact of mutation scale under varying crossover rates and population sizes.

Among these parameters, population size had the most consistent and noticeable effect on FV in the IEMA. Larger populations (e.g., 350) maintained variance more steadily across generations, regardless of mutation scale or crossover rate. In contrast, smaller populations (e.g., 100) showed a sharper and faster decline in variance. This suggests that, within the IEMA setup, population size plays a stabilizing role in maintaining parametric diversity.

5.2.2 Phylogenetic Space

This section presents the results regarding the metrics focusing on the phylogenetic space.

Pairwise Distance

Higher crossover rates led to deeper evolutionary trajectories by applying greater reproductive pressure on the population. With more recombination occurring at each generation, individuals inherit traits from multiple ancestors over time therefore extend the depth of the evolutionary trajectory. This effect was captured by the PD metric, which generally increased in response to higher crossover rates.

As shown in Figure 5.5, this trend is most evident in larger populations (e.g., 200 and 350) showing consistently higher PD at increased crossover rates. As the population gets smaller, the effect of crossover rate on PD became less distinct. This suggests that, in the IEMA system, the impact of crossover on evolutionary depth is weaker when population size is limited.

Larger populations maintained shorter evolutionary distances between individuals due to the continuous introduction of new individuals at each generation. Since larger populations require greater amount of new individuals to sustain their size, the evolutionary trajectory expands more horizontally rather than vertically.

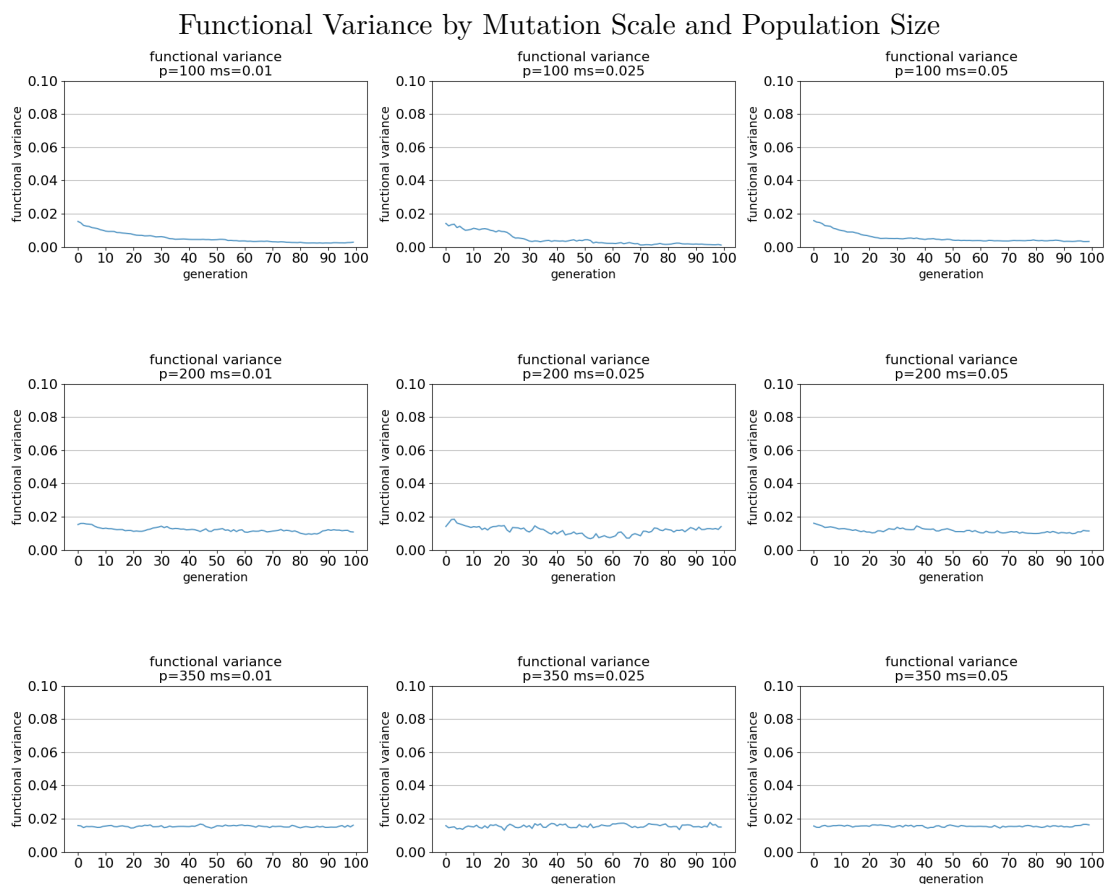


Figure 5.4: Functional Variance (FV) over generations on varying mutation scales and population sizes, for crossover rate 0.65. Rows are population sizes, columns are mutation scales. x-axis is generation number and y-axis is functional variance value.

The results for the PD metric confirmed this effect by showing a clear inverse relationship with population size. This shows that the metric is effective in capturing how population size influences the depth and spread of lineages in the evolutionary history.

As shown in Figure 5.6, PD did not show a significant change in response to varying mutation scale, which aligns with expectations. In the IEMA system, mutation is applied as a parametric shift within the behavior space, and does not alter the structure of the evolutionary tree in a way that increases lineage depth. Since mutation operates on individuals after selection and crossover, its effect is primarily spatial rather than historical.

The observed trends confirm that the PD metric captured the aforementioned evolutionary dynamics. Its responsiveness to both population size and crossover rate demonstrated its usefulness in measuring the structural depth and divergence within the evolutionary space. This suggests that PD can serve as an indicator of how broadly individuals are distributed across evolutionary trajectories and how evolutionary history unfolds in terms of maintaining diverse lineages versus

Pairwise Distance by Crossover Rate and Population Size

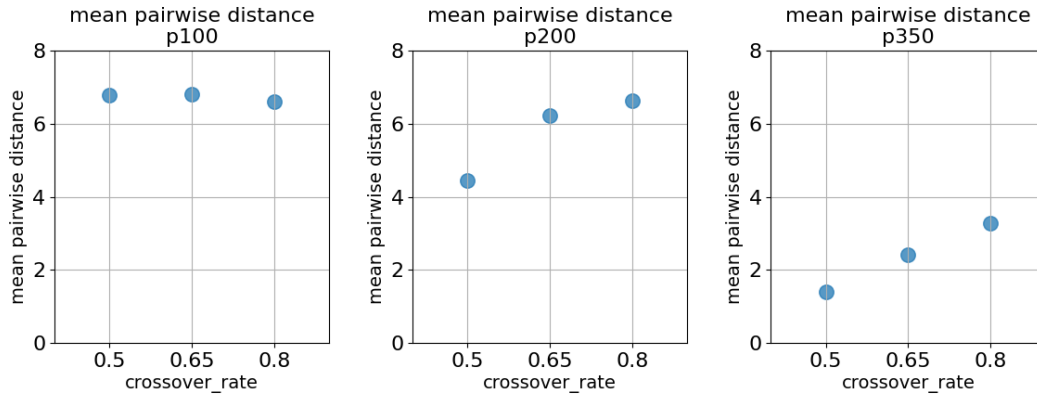


Figure 5.5: Pairwise Distance (PD) on varying crossover rates. The y-axis represents the PD, and the x-axis represents generations

Pairwise Distance by Mutation Scale and Population Size

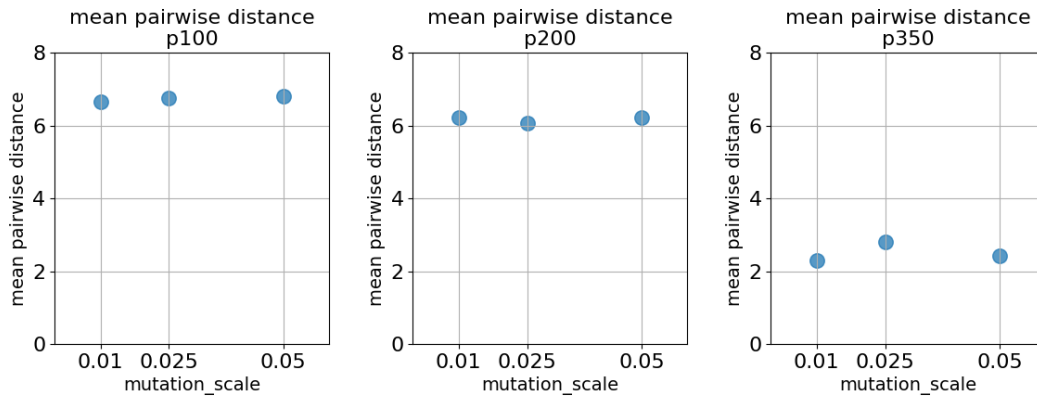


Figure 5.6: Pairwise Distance (PD) on varying mutation scales. The y-axis represents the PD, and the x-axis represents generations

converging toward similar solutions over successive generations.

Additionally, in the smallest population size of 100, an exception to the general trend was observed, where PD did not consistently increase with crossover rate and instead showed a slight decrease. This can be explained by the fact that, at lower population sizes, the introduction of new individuals is extremely limited or even nonexistent. In this case, crossover determines the structure of evolutionary lineages, as it dictates how many individuals are recombined rather than newly introduced. With higher crossover rates, more individuals undergo recombination. This results in forming more connections within the existing set of ancestors, which led to a few dominant lineages persisting. This behavior ultimately shortens the average pairwise distances instead of increasing them.

In PD calculation, averaging the pairwise distances across the entire population led to a loss of information regarding the distribution of distances. A more detailed analysis of the distribution of pairwise distances within different population subsets

Root Contribution Index by Crossover Rate and Population Size

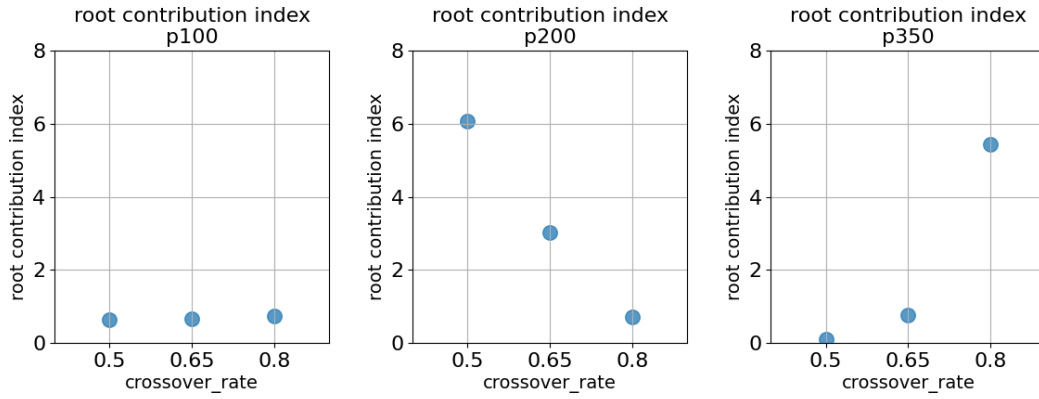


Figure 5.7: Root Contribution Index (RCI) with varying crossover rates under different population sizes (p).

could provide a more nuanced understanding of the evolutionary structure.

Root Contribution Index

For a population size of 100, the Root Contribution Index (RCI) remained stable across varying crossover rates. For a population size of 200, however, the RCI decreased drastically as the crossover rate increased. In contrast, for a population size of 350, the RCI increased with higher crossover rates.

The RCI varied across different population sizes, but there was no consistent pattern. The highest RCI was observed for a population size of 200, while the lowest was for 350.

For mutation scale, no significant trend was observed. This result aligns with the expectation that mutation primarily affects the parametric space, whereas the RCI focuses on the phylogenetic space. As mutation occurs in the parametric space at the last stage among evolutionary operations, it is not expected to have a significant impact on the RCI.

Overall, the RCI results showed limited variation across different experimental setups. While population size and crossover rate did impact the RCI to some extent, the effect was not consistent. Figures 5.7 and 5.8 show the results regarding RCI.

5.2.3 Categorical Space

This section presents the results regarding the metrics in the categorical space.

Categorical Diversity

The results indicate that Categorical Diversity (CD) in the ontological space is highly influenced by the balance between crossover rate and population size. When the crossover-to-population size ratio is low, which means that a relatively large population is evolving under a less aggressive crossover rate, diversity is better preserved throughout generations. Conversely, when the crossover rate is high

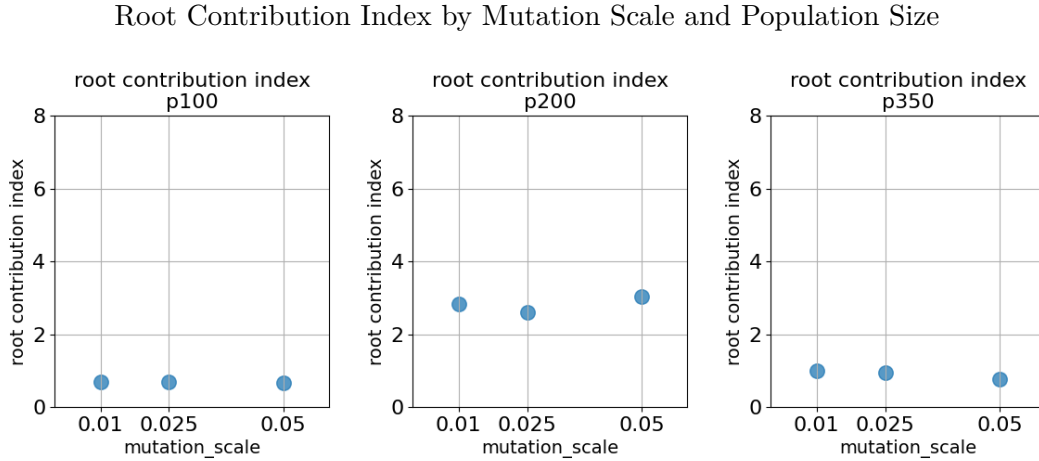


Figure 5.8: Root Contribution Index (RCI) with varying mutation scales under different population sizes (p).

relative to the population size, diversity declines more rapidly, indicating stronger convergence. This trend is particularly evident in lower population sizes where diversity drops steeply early in the evolutionary process. For larger population sizes, however, the effect of crossover rate appears to diminish with CD stabilizing across generations, regardless of the crossover rate. This suggests that, at this scale, the evolutionary process does not significantly alter CD over generations.

In higher population sizes (e.g., population sizes 350 and 500), once CD stabilizes over generations, the stabilized values tend to be lower compared to those in smaller populations. This could be due to the fact that the number of categories explored reaches a saturation point, meaning that most of the available categorical space has already been covered. And since CD is normalized by population size in its formulation (see Section 4.2.3), larger populations inherently lead to lower relative diversity values, even when the absolute number of explored categories remains high.

The CD metric was sensitive to the differences caused by varying crossover rates and population sizes. The sharp declines seen at higher crossover rates and lower population sizes confirm that the metric is successful in measuring categorical convergence in the ontological space. Figure 5.9 shows the CD for each configuration.

CD remained relatively stable across mutation scales, with only minor differences. While one might expect higher mutation scales to increase categorical spread by introducing more variation, the results show that beyond a certain point, increasing mutation scale does not significantly impact categorical diversity.

In smaller populations (e.g., 100), diversity dropped over generations for low mutation scales (0.01), while moderate scales (0.025–0.05) helped slow this decline. In larger populations (e.g., 200 and 350), the diversity was more consistent regardless of mutation scale, showing that population size played a stronger role in maintaining CD than mutation strength alone. Figure 5.10 presents CD across varying mutation scales for different population size.

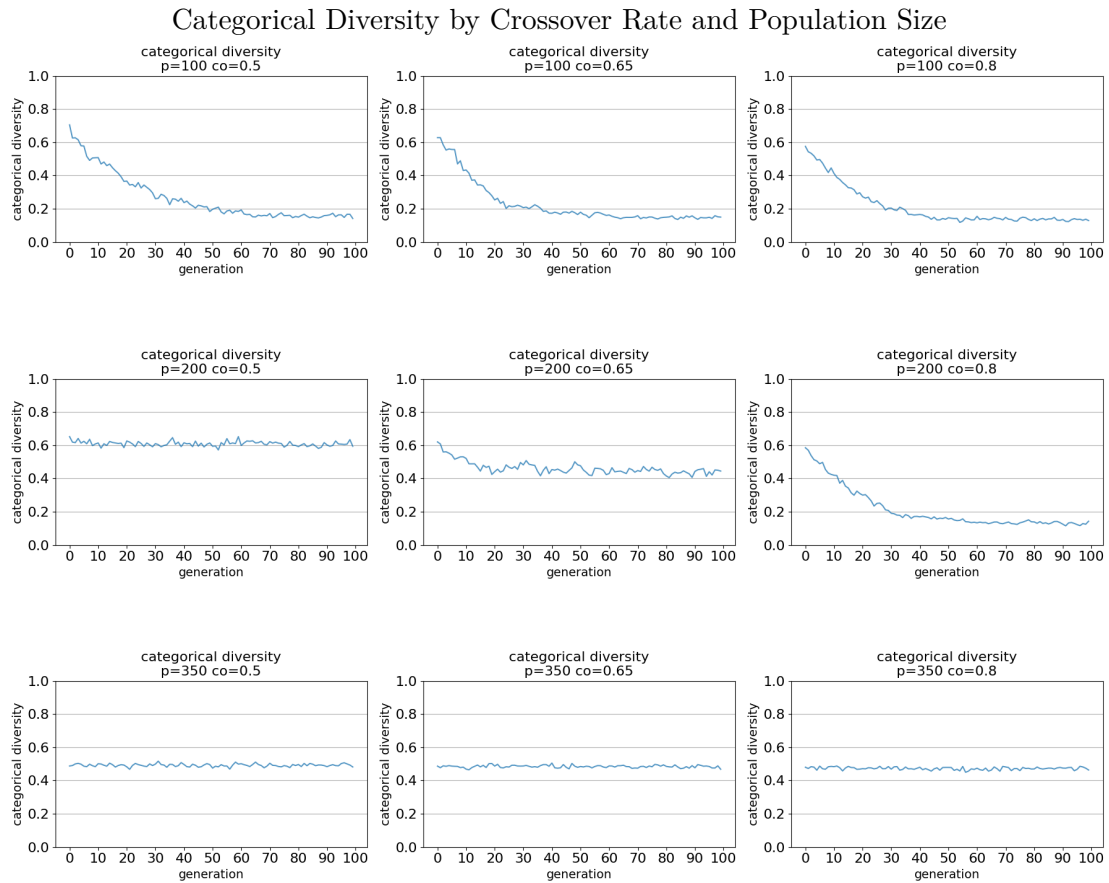


Figure 5.9: Categorical Diversity (CD) over generations with varying crossover rates. Rows represent different population sizes (p), while columns represent different crossover rates (co). The y-axis represents the categorical diversity, and the x-axis represents generation number.

Category Change Rate

The results for the Category Change Rate (CCR) revealed fluctuation patterns over generations, particularly influenced by the crossover-to-population size ratio. As shown in Figure 5.11, when the crossover rate is high and the population size is low, it is possible to see slight oscillations in the CCR over generations. This pattern likely arises because recombination periodically disrupts convergence by introducing new individuals. However, as selection reinforces similarity, the population stabilizes until the next wave of recombination and decreases CCR. The higher the crossover rate was, the more frequently these fluctuations occurred within 100 generations.

In contrast, at lower crossover rates and larger population sizes, CCR remains more stable, with fewer abrupt changes. This suggests that the categories remain changing with introduction of new individuals or lack of concentration with reproduction.

The overall magnitude of CCR was notably higher in smaller population sizes, showing more pronounced drops and rises. This suggests that when the number of individuals is low or the crossover pressure is high, category changes occur faster, as each new reproduction has a greater impact on the overall distribution.

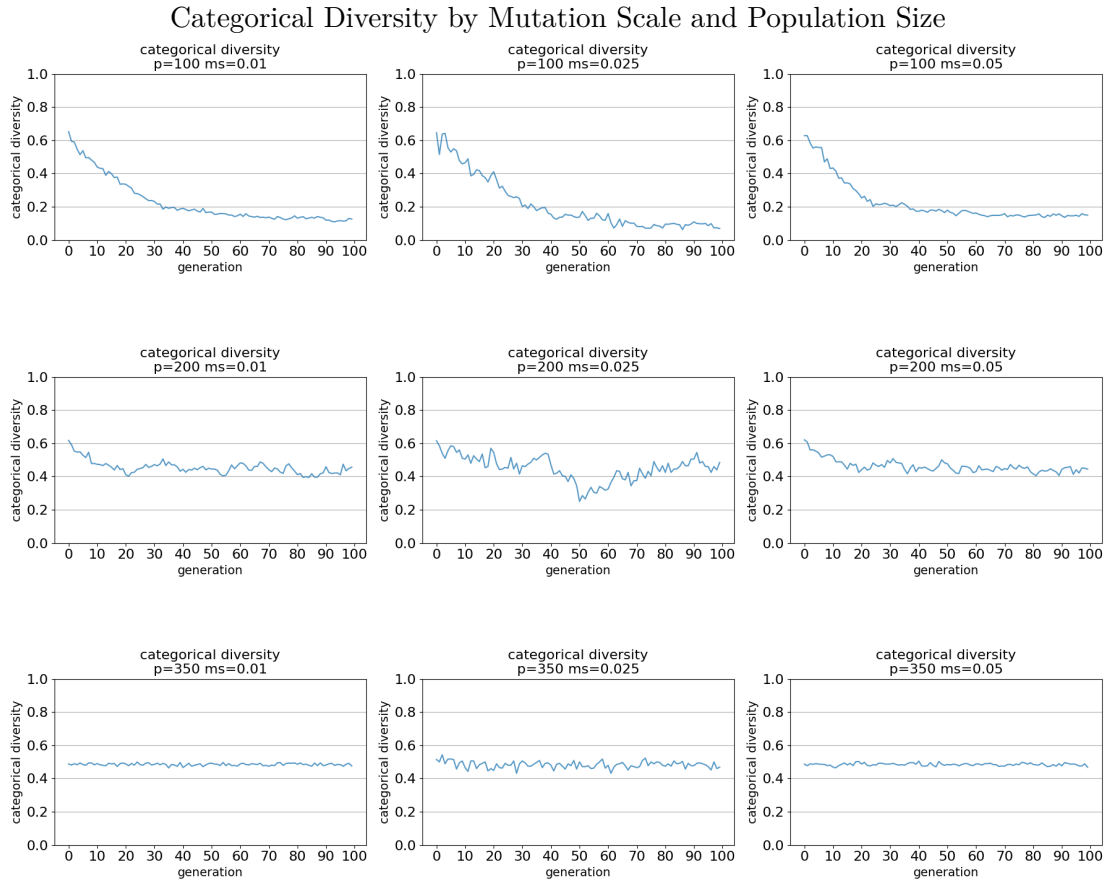


Figure 5.10: Categorical Diversity (CD) over generations with varying mutation scale. Rows represent different population sizes (p), while columns represent different mutation scales (ms). The y-axis represents the categorical diversity, the x-axis represents generations

As shown in Figure 5.12, CCR did not show strong variation in response to different mutation scales. Across all population sizes, the overall trend remained relatively stable regardless of the change in mutation scale. Interestingly, at the intermediate scale (0.05), a slightly more noticeable decrease was observed, though the reason for this is not entirely clear. One possible explanation is that higher mutation scales may cause parameter values to exceed useful regions of the space, either hitting boundaries or jumping to areas that map back to similar categories, therefore not contributing to meaningful change. Meanwhile, the intermediate mutation scale might have been just enough to push individuals into new areas while still staying within relevant bounds. In any case, the results suggest that, in IEMA, simply increasing mutation scale did not guarantee higher rate of change in the Audio Set categories.

5.2.4 Data and Category Coverage

As detailed in Section 4.2.4, coverage is evaluated under Dataset Coverage (DC) and Category Coverage (CC).

Across all population sizes, higher crossover rates generally led to slightly



Figure 5.11: Category Change Rate (CCR) over generations with varying crossover rates. Rows represent different population sizes (p), while columns represent different crossover rate (co). The y-axis represents the CCR, the x-axis represents generation number.

lower coverage. This suggests that stronger crossover pushes the system toward recombination of existing individuals rather than discovering new ones from the dataset. In IEMA, this effect is even more pronounced because higher crossover rates result in more offspring being generated through recombination, which reduces the need to sample new individuals from the corpus to fill the population. As a result, the system leans more on recombination and less on bringing in novel data, which limits overall dataset coverage.

Dataset coverage increased clearly with population size, regardless of the crossover rate. For population size 100, coverage remained very low across all crossover values, around 7%. At higher population sizes (e.g., 200 and 350), coverage improved but still dropped as crossover rate increased. The highest coverage was achieved with population size 350, where nearly half of the dataset was accessed.

When the crossover rate was fixed and mutation scale varied, dataset coverage remained largely unchanged. The extent of mutation did not significantly affect the number of new samples explored, as even with higher mutation, the system mapped to similar regions of the corpus. Instead, population size had a consistent impact: larger populations resulted in higher dataset coverage, regardless of mutation



Figure 5.12: Category Change Rate (CCR) over generations with varying mutation scale. Rows represent different population sizes (p), while columns represent different mutation scales (ms). The y-axis represents the CCR, the x-axis represents generations

strength. This suggests that population size, rather than mutation alone, was the key factor in reaching more of the dataset. DC plots with varying parameters are shown in Figure 5.13. *Ideal maximum coverage* represents the theoretical upper bound based on population size.

Category Coverage (CC) increased with population size across all crossover rates. In the smallest population (100), coverage remained below 80%, with only a slight increase as crossover rate increased. For population size 200, coverage reached up to the high 90% range at the lowest crossover rate (0.5), but then noticeably dropped as crossover increased. In the largest population (350), category coverage was consistently close to 100% and remained stable regardless of crossover rate.

Category coverage slightly increases with crossover rate in the smallest population (100), but decreases in the medium population (200). This may be due to the fact that, in smaller populations, crossover helps diversify the limited gene pool and reach more categories. In contrast, with population size 200, the system already accesses a wide range of categories at low crossover. As crossover rate increases, more individuals are generated from within the population, reducing the need to sample new data. This internal recycling appeared to limit further

Dataset Coverage by Crossover Rate and Mutation Scale

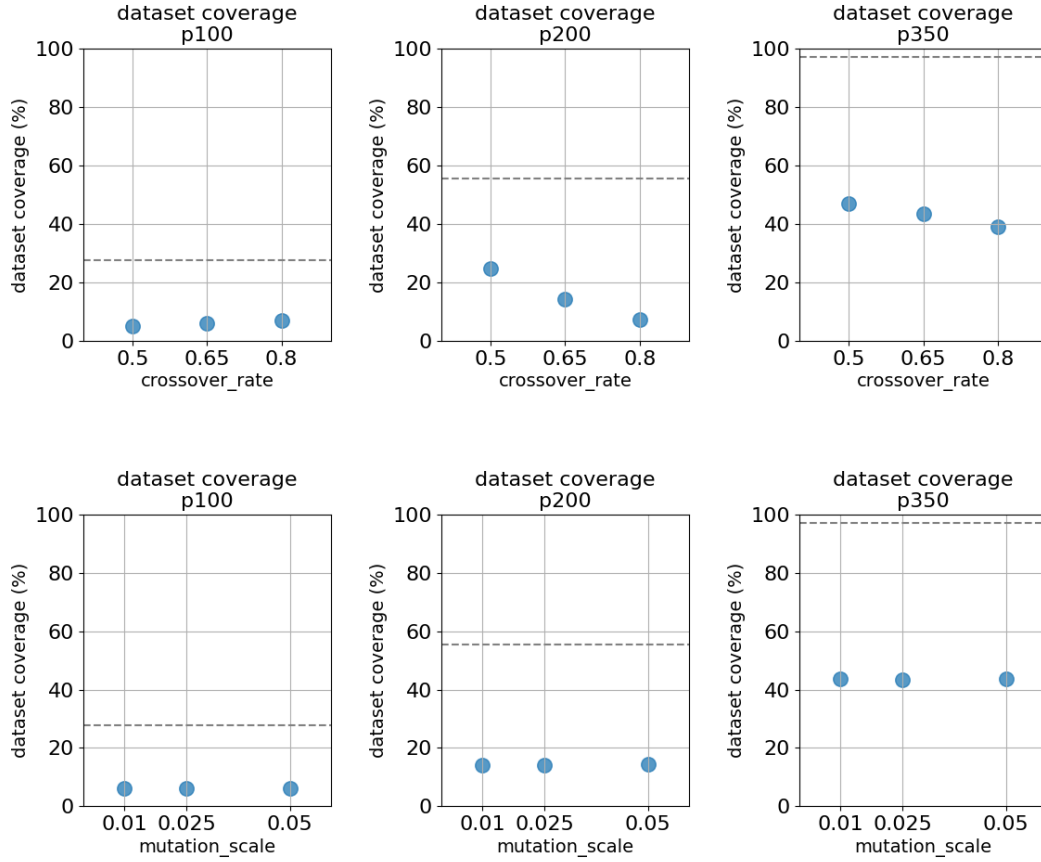


Figure 5.13: Dataset Coverage (DC) over generations with varying crossover rates (top) and mutation scales (bottom), for different population sizes. The dashed line indicates the *ideal maximum coverage* in 100 iterations for given population size.

category discovery, causing the coverage to drop.

Category Coverage did not show any significant or consistent pattern in response to changes in mutation scale. Across all population sizes, it remained relatively stable regardless of how much the mutation scale was increased. This suggests that mutation strength, under the current IEMA setup, did not substantially influence how many unique categories are reached. The population size continued to have a more noticeable effect, with higher sizes naturally leading to higher and more stable coverage overall. Figure 5.14 shows Category Coverage with varying crossover rates and mutation scales.

Across all the evaluated metrics, a key structural factor influencing the system's behavior was the interplay between crossover rate and population size. This relationship was a direct result of how crossover operates within the evolutionary process. The crossover rate defines the proportion of the population that undergoes recombination in each generation. Since crossover is applied to a proportion of the population rather than a fixed number of individuals, its effect scales with population size. This dependency arises specifically from the nature of crossover as a population-wide operation.

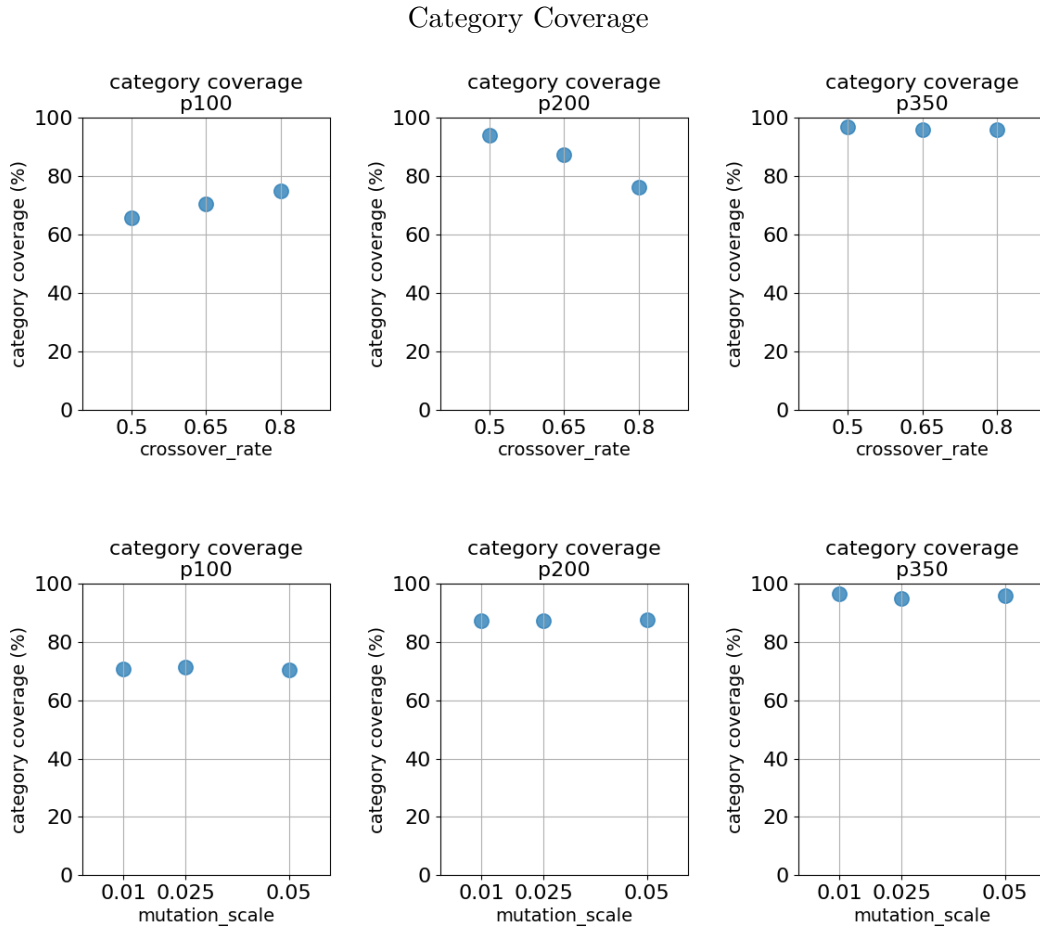


Figure 5.14: Category Coverage (CC) with varying crossover rates (top) and mutation scales (bottom), for different population sizes. The dashed line indicates the *ideal maximum coverage* in 100 iterations for given population size.

Mutation scale, in contrast, operates at the individual level and did not show a strong or consistent impact on most of the evaluated metrics. This suggests that, within the IEMA setup, mutation alone may not be sufficient to drive significant exploration of the corpus unless carefully balanced with other system dynamics.

5.3 Targeted System Behaviors

Based on the results across crossover rate, mutation scale and population size, several configurations were selected to represent different desired system behaviors, as explained in Section 3.3.3. The selection was guided by how the evaluation metrics responded to different parameter setups. The selected configurations satisfying the three scenarios are given in Table 5.1.

These selected configurations were then run in isolation for a more focused evaluation. As explained in Section 3.3.4, two further tests were conducted on top of each configuration: one varying the dimensionality reduction technique (e.g., PCA or t-SNE), and another toggling the Novel Selection mechanism enabled or disabled. These parameters are introduced and discussed in Section 3.3.4.

Table 5.1: Selected values for different scenarios

	Population Size	Crossover Rate	Mutation Scale
Scenario 1	150	0.5	0.01
Scenario 2	200	0.8	0.05
Scenario 3	200	0.5	0.05

5.3.1 Comparison of Dimensionality Reduction Methods

Figures 5.15, 5.16, 5.17 and 5.18 show results for the three scenarios under varying dimensionality reduction methods.

Across all three scenarios, the choice of dimensionality reduction algorithm had varying degrees of influence on the system's behavior. In Scenario 1, t-SNE generally resulted in slightly higher categorical diversity and category change rate, and these metrics began to fluctuate earlier in the process. This suggests that t-SNE may promote a more scattered spread in the projected space. However, other metrics remained highly similar between the two methods.

In Scenario 2, t-SNE led to a faster decrease in Functional Variance, which indicates quicker local convergence in the parametric space. While both methods produced similar overall categorical diversity, t-SNE exhibited more local fluctuations, whereas PCA remained more stable and gradually increased toward the end. This could point to t-SNE favoring short-term variation, while PCA maintained steadier progress. Other metrics remained largely unchanged.

In Scenario 3, there was little difference between the methods in terms of Functional Variance, categorical diversity, or category change rate. However, t-SNE produced higher pairwise distances, suggesting deeper evolutionary trajectories or more varied ancestral paths in the phylogenetic space. Coverage values, both in terms of data and categories, did not show significant differences in response to the projection method in any of the scenarios.

Overall, while the projection method subtly affected the dynamics of evolution over time, the differences were not drastic. This suggests that under reasonably distinct dimensionality reduction techniques like PCA and t-SNE, the IEMA system remains relatively stable in its behavior.

5.3.2 Impact of Novel Selection Mechanism

Figures 5.19, 5.20, 5.21 and 5.22 show results for the three scenarios under varying Novel Selection configurations.

In the final set of tests, the effect of the Novel Selection mechanism was evaluated by running each configuration with Novel Selection both enabled and disabled. This mechanism is explained in detail in Section 3.3.4.

Overall, novel selection showed clear influence in the parametric and ontological spaces over time. In the phylogenetic space, however, the metrics were less affected, as Novel Selection mechanism primarily operates at the level of individual sampling and does not significantly alter the underlying evolutionary lineage.

For Scenario 1, in the parametric space, functional variance remained similarly low across both conditions with no major impact caused by Novel Selection. In the ontological space, both categorical diversity and category change rate were consistently higher with Novel Selection enabled. This suggests that not allowing

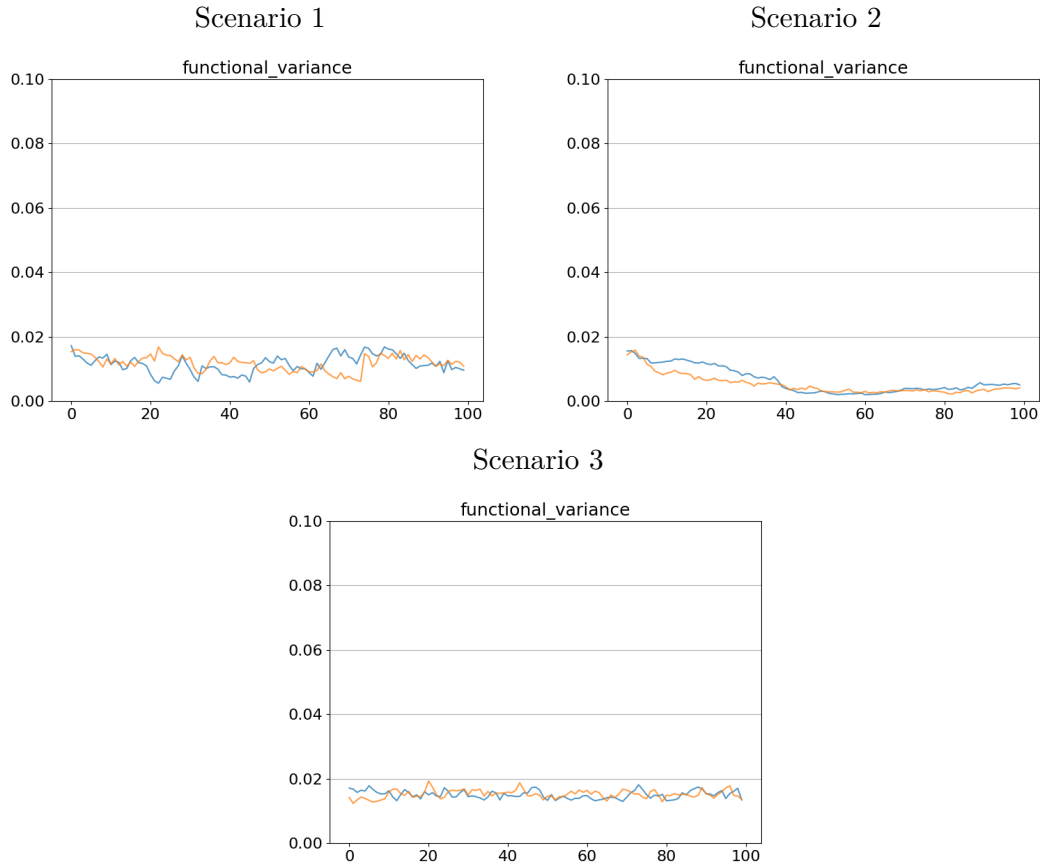


Figure 5.15: Functional Variance over generations, under varying projection methods and scenarios. Blue line shows PCA, while orange line shows t-SNE.

repeated samples from the corpus helped maintain more varied and dynamic populations across generations. In terms of coverage, dataset and category coverage saw a large increase with novel selection. This shows that more unique samples and categories were accessed when repetitions were avoided.

For Scenario 2, in the parametric space, Functional Variance declined more quickly when Novel Selection was enabled, followed by a slight increase in later generations. In contrast, when Novel Selection was off, both the initial convergence and later increase were more gradual. In the ontological space, categorical diversity and category change rate were consistently higher under Novel Selection, with a sharp increase in later generations. This indicates that novelty in corpus sampling contributed to more dynamic and varied categorical behavior over time. Dataset Coverage nearly doubled with novel selection, which confirms its strong role in expanding the explored sample space, and Category Coverage increased in parallel.

Finally, for Scenario 3, Novel Selection had a relatively limited effect across most metrics. In the parametric space, Functional Variance remained low and stable in both conditions. Ontologically, the Category Change Rate and Categorical Diversity remained high and stable in both conditions, but showed slight gains under Novel Selection. Dataset Coverage was again significantly improved by Novel Selection, though Category Coverage was already saturated and remained almost unchanged.

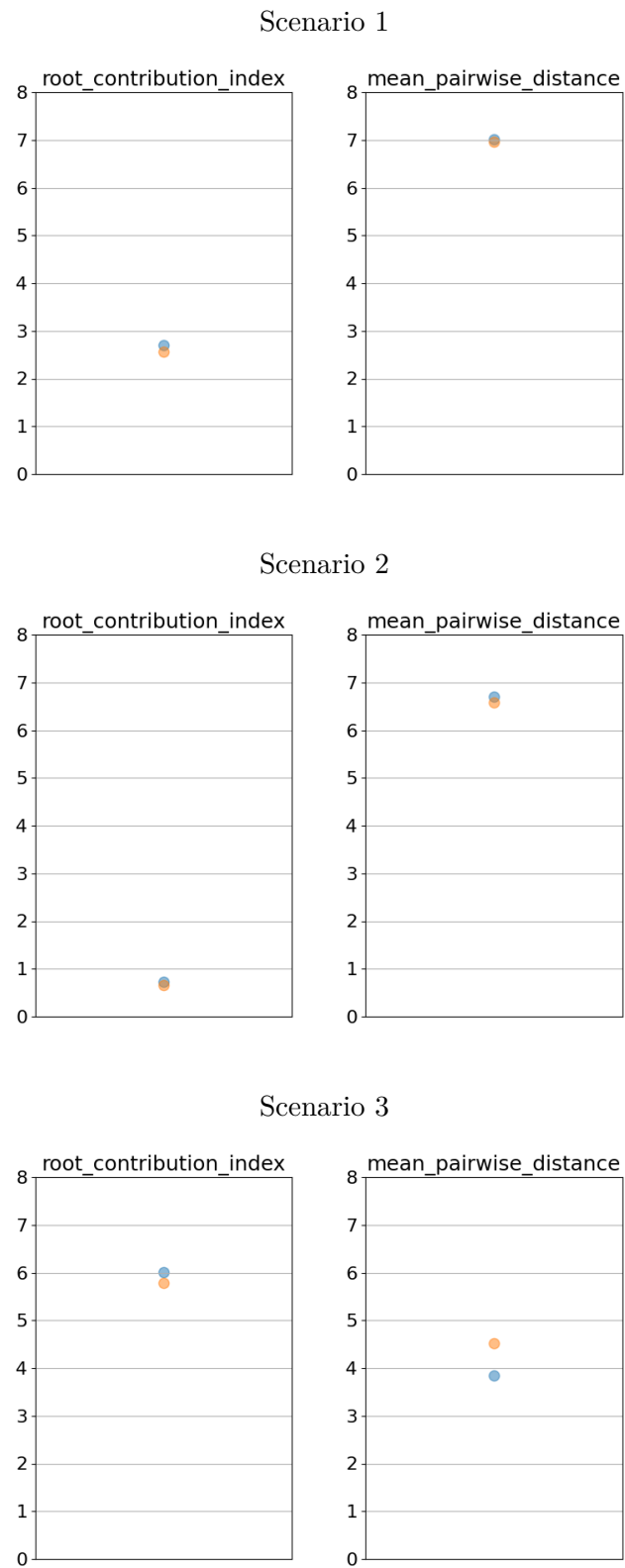


Figure 5.16: Phylogenetic Space metrics (RCI & PD) over generations, under varying projection methods and scenarios. Blue dot shows PCA, while orange dot shows t-SNE.

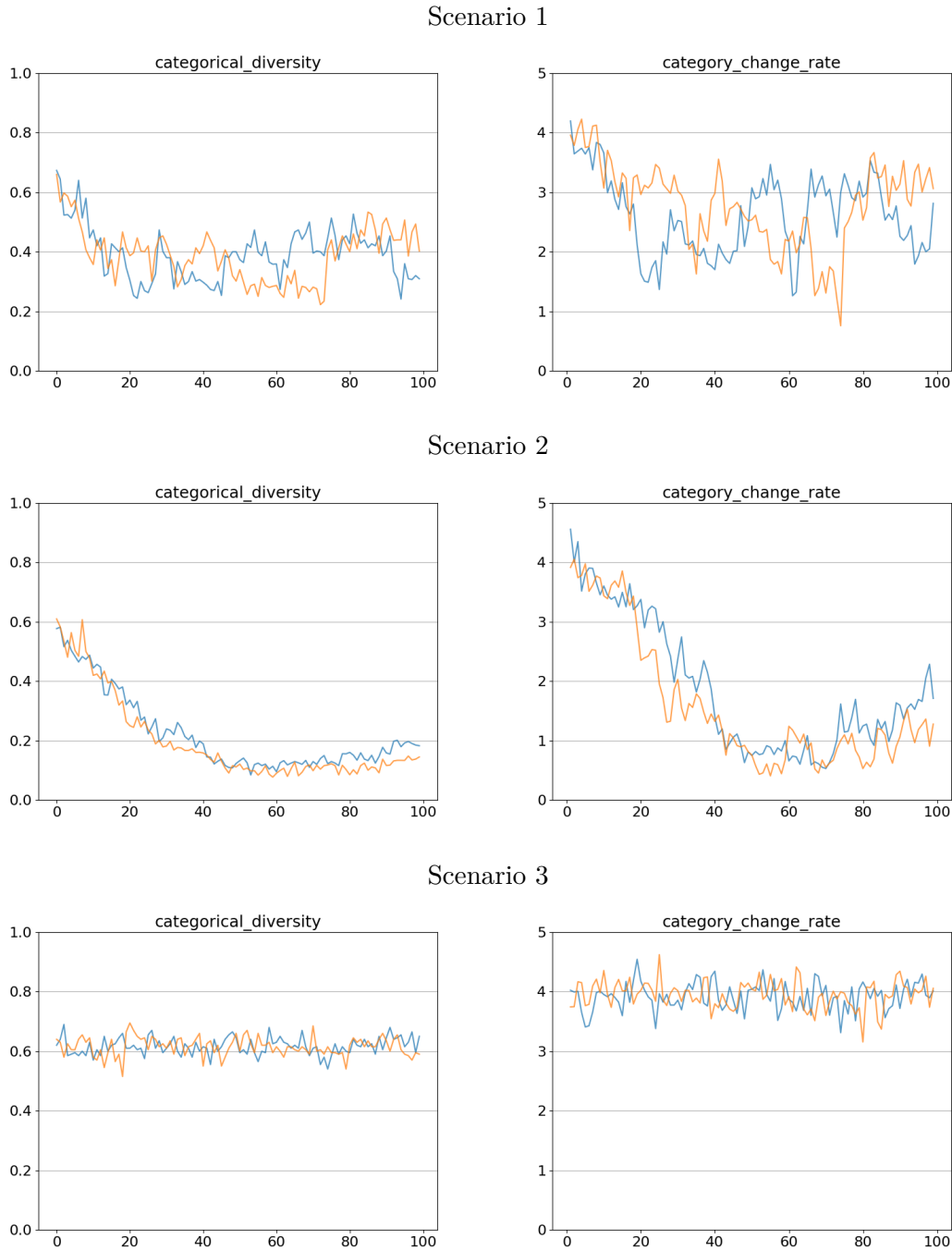


Figure 5.17: Categorical Space metrics (CD & CCR) over generations, under varying projection methods and scenarios. Blue line shows PCA, while orange line shows t-SNE.

Overall, the Novel Selection mechanism significantly increased both diversity and novelty-related metrics in the parametric and categorical spaces. For the given scenarios, it led to broader exploration and higher functional variation in some scenarios, as well as a more dynamic categorical behavior. Under Novel Selection, many more combinations of parameters could potentially lead to high-diversity or high-novelty outcomes. In the phylogenetic space, however, Novel Selection had minimal impact, which is expected, as it does not directly influence

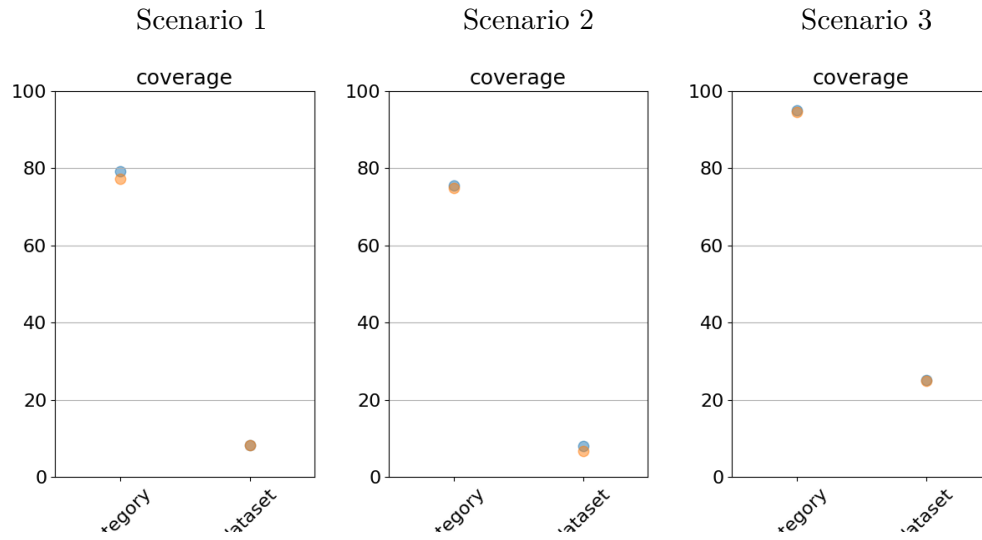


Figure 5.18: Coverage metrics (DC & CC) over generations, under varying projection methods and scenarios. Blue dot shows PCA, while orange dot shows t-SNE.

the ancestral lineage of individuals. Instead, its influence is most visible in the immediate composition of the population and its evolving characteristics.

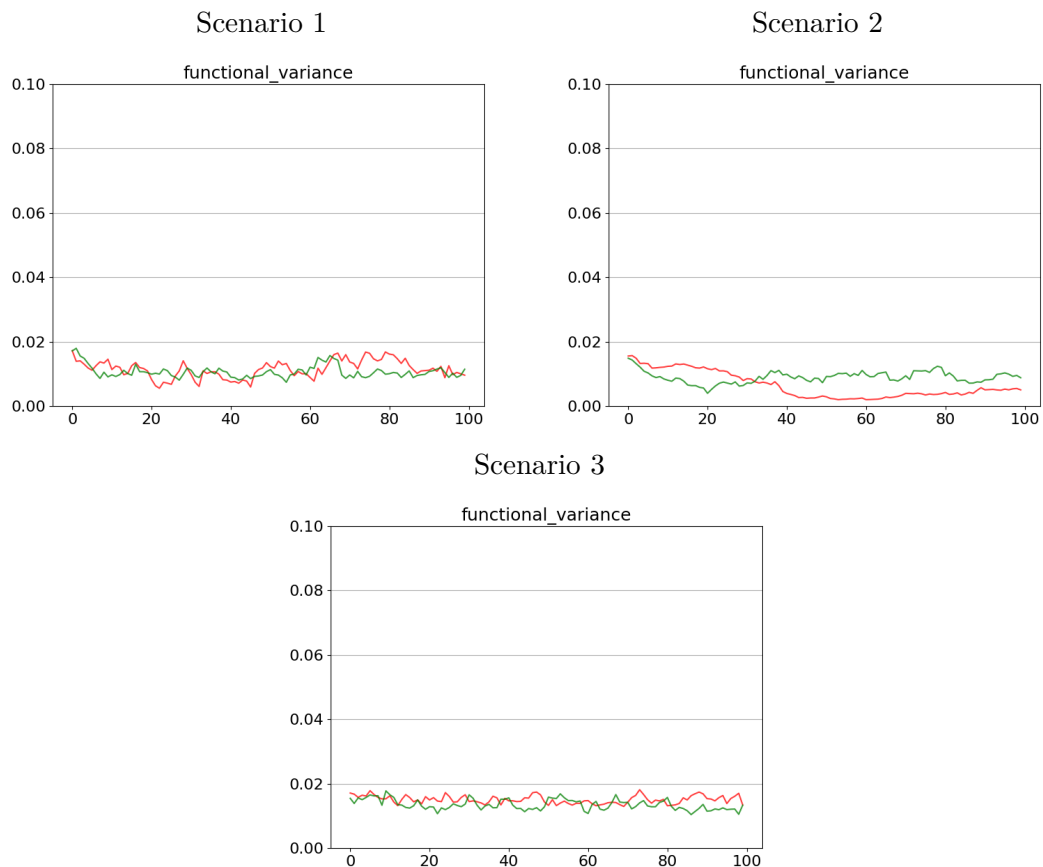
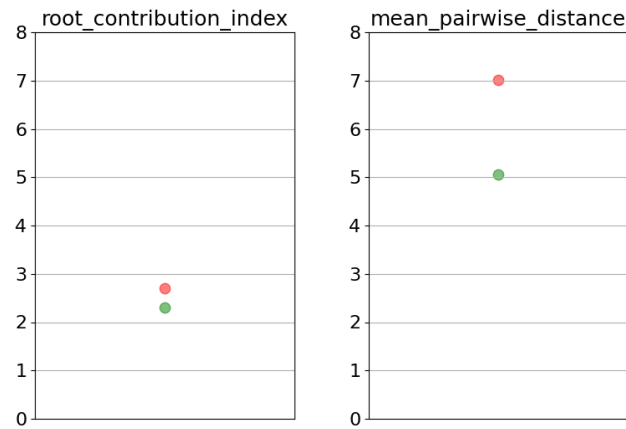
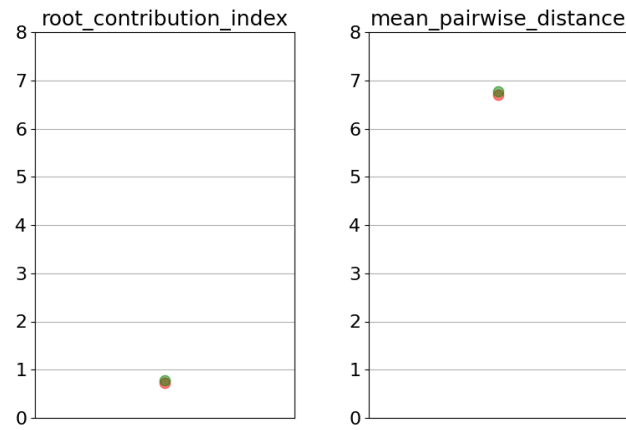


Figure 5.19: Functional Variance across generations for different Novel Selection configurations and scenarios. Green line represents Novel Selection enabled, while red line represents disabled.

Scenario 1



Scenario 2



Scenario 3

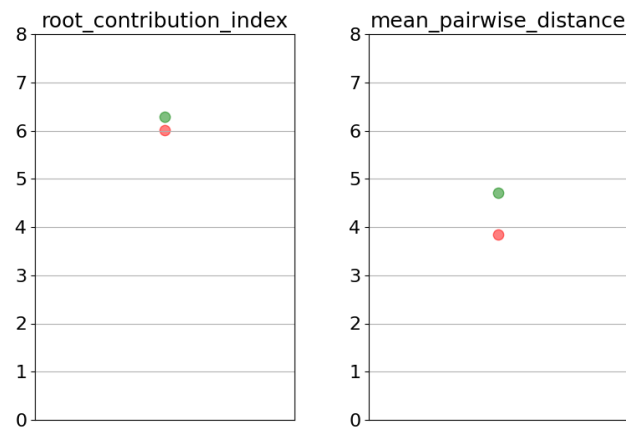


Figure 5.20: Phylogenetic Space metrics (RCI & PD) across generations for different Novel Selection configurations and scenarios. Green dot represents Novel Selection enabled, while red dot represents disabled.

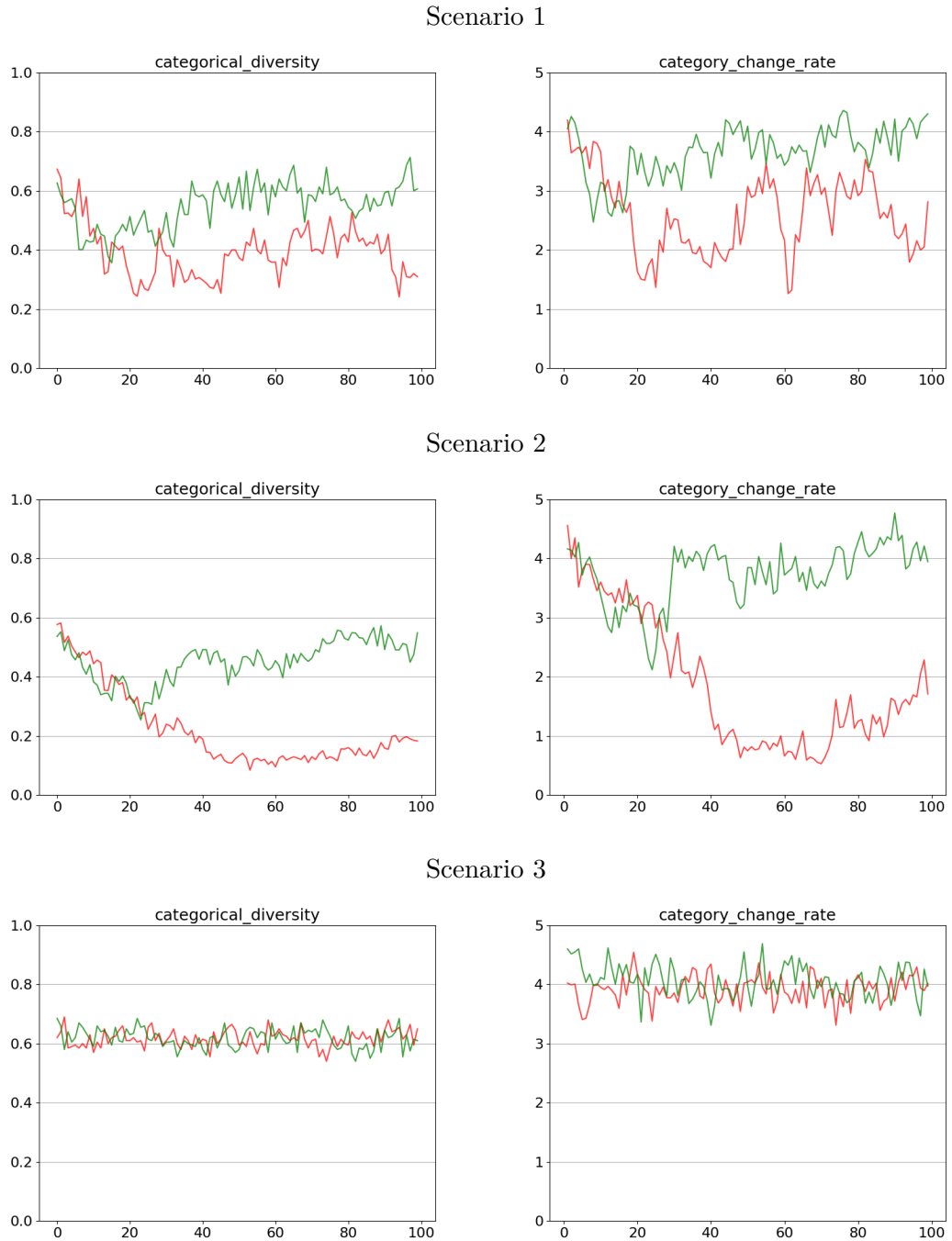


Figure 5.21: Categorical Space metrics (CD & CCR) across generations for different Novel Selection configurations and scenarios. Green line represents Novel Selection enabled, while red line represents disabled.

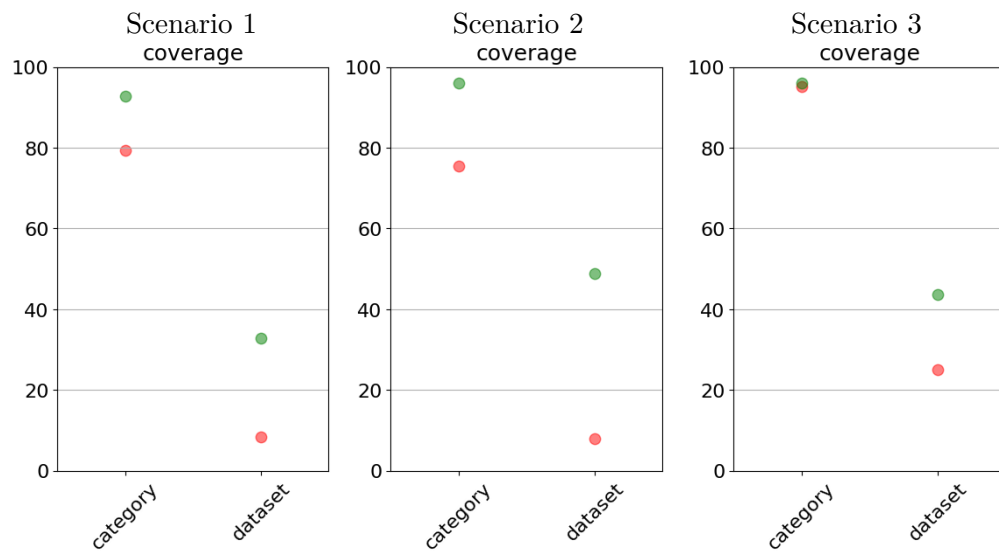


Figure 5.22: Coverage metrics (DC & CC) across generations for different Novel Selection configurations and scenarios. Green dot represents Novel Selection enabled, while red dot represents disabled.

Chapter 6

Conclusion

This thesis presented the development and evaluation of the Immersive Evolutionary Motion-controlled Algorithm (IEMA), an interactive system that integrates Interactive Evolutionary Algorithms (IEAs) with discrete data, body motion interaction, and one-class feedback mechanisms. Evolution was primarily driven by IEA, while three main methods were utilized for user interaction: motion capture, dimensionality reduction, and spatial audio. For evaluation of such system, a set of mathematical measures were introduced, which informed the design and implementation of such interactive computer music system.

The results validated the approach for facilitating the interactivity in the proposed system. The research question regarding how an interactive music system can be designed to facilitate three-dimensional navigation and feedback-driven exploration of a high-dimensional discrete space is explored in Chapters 3. The findings answer the research question, and support the feasibility of designing the proposed system.

Moreover, the research question about designing and implementing such an algorithm is answered in Chapters 3, where the orchestration of IGAs, dimensionality reduction, and unary feedback mechanisms is demonstrated to effectively facilitate the system's functionality. The results obtained confirm the feasibility of the proposed algorithm for the interactive evolution of discrete audio data in this setup. This establishes that the algorithm can be successfully developed to meet the requirements.

Finally, the results shed light on how the proposed system can be effectively evaluated to inform design and implementation. Therefore, the research question regarding the development and application of evaluation metrics for understanding the system's behavior is addressed, where the introduced metrics were shown to provide meaningful insights into the system's exploration and evolutionary dynamics.

Overall, the results obtained from this research demonstrated the feasibility of designing an interactive music system around evolutionary algorithms, that facilitates three-dimensional navigation and feedback-driven exploration of a high-dimensional discrete space. The thesis did not aim to prove the optimality of the system but instead offered iterative stage-wise prototypes to explore how such a system can be built and how the design choices affect the overall interaction and evolutionary process.

While this research focused on a systematic user simulation, where user behavior was abstracted within a narrower scope, the results would benefit from being tested with real users. Such analysis would provide deeper insights into the interaction design and algorithmic behavior, thereby enhancing the understanding of how users engage with the evolving soundscapes and what algorithmic necessities are introduced with real users to make it user-centric more maybe.

Additionally, testing the system with different audio datasets could further assess its robustness and introduce interesting variations. Moreover, IEMA has significant potential for exploration in various domains beyond audio. Future work could extend the IEMA framework to handle different data, which would provide insights into its adaptability across different domains. Finally, this research is limited to working with discrete data, which imposes certain constraints. Comparing the system's behavior with continuous parametric spaces could offer valuable insights.

In the proposed IEMA setup, the evolutionary process takes considerable time to produce meaningful results, which may not be ideal for applications requiring rapid, satisfactory outcomes for users. However, the IEMA can be more suitable for long-term installations, where continuous evolution over extended periods can provide progressively refined soundscapes.

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